

Asymptotically flat spacetimes represent ideally isolated systems in GR.

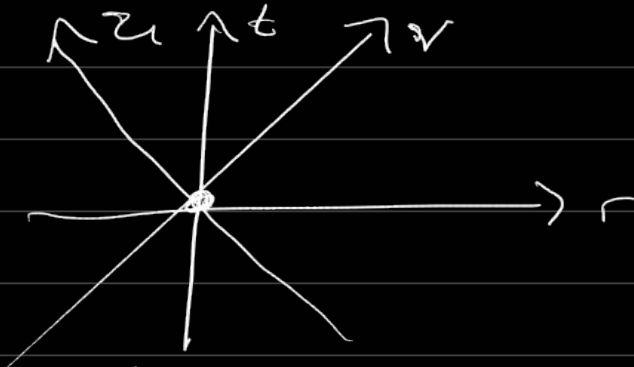
problem: No longer can generalise/have a background flat metric η in terms of which the fall off rates of the curvature of g can be specified.

$\rightarrow$ solⁿ: find a "global" coordinate system in which $g_{\mu\nu} \sim \eta_{\mu\nu} + \mathcal{O}(\frac{1}{r})$ as $r \rightarrow \infty$ along a spatial or null direction and r moves $r \in \mathbb{R}^3 : r^2 = (x^1)^2 + (x^2)^2 + (x^3)^2$

Conformal Infinity:

M^4 in spherical: $ds^2 = -dt^2 + dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$

• Radiation from a massless scalar field ϕ



$$\begin{cases} u = t - r \\ v = t + r \end{cases} \quad \left| \quad \begin{aligned} \frac{u+v}{2} &= t \\ \frac{v-u}{2} &= r \end{aligned} \right.$$

$$\Rightarrow ds^2 = -du dv + \frac{1}{4}(v-u)^2(d\theta^2 + \sin^2\theta d\phi^2)$$

* Outgoing radiation with u fixed: $v \rightarrow \infty$ from ϕ .
 $ds^2 \rightarrow \frac{1}{4}(v-u)^2(d\theta^2 + \sin^2\theta d\phi^2)$

• Look at radiation "to the right"

$v \rightarrow \infty$ from ϕ

• determined from " $\frac{1}{v}$ piece" of ϕ in this limit:

$$\square \phi = 0$$

Laplacian in spherical coords ∂^2
 $\square = \frac{1}{r^2} \partial_r (r^2 \partial_r) + \frac{1}{r^2} \partial_{\theta} (r^2 \sin\theta \partial_{\theta}) + \frac{1}{r^2 \sin\theta} \partial_{\phi}^2$

$$+ \frac{1}{r^2} \sin(\alpha) \frac{\partial}{\partial \phi} (\sin(\alpha) \frac{\partial}{\partial \phi})$$

$$\Rightarrow \frac{\partial}{\partial r} = \frac{\partial u}{\partial r} \frac{\partial}{\partial u} + \frac{\partial v}{\partial r} \frac{\partial}{\partial v} = -\partial_u + \partial_v$$

$$\frac{\partial}{\partial t} = \frac{\partial u}{\partial t} \frac{\partial}{\partial u} + \frac{\partial v}{\partial t} \frac{\partial}{\partial v} = \partial_u + \partial_v$$

$$\square = -\partial_t^2 + \nabla^2$$

$$= (\partial_u + \partial_v)(\partial_u + \partial_v) + \nabla^2 = \partial_u^2 + 2\partial_v \partial_u + \partial_v^2 + \nabla^2$$

$$= \partial_u^2 + 2\partial_v \partial_u + \partial_v^2 +$$

$$r^2 = \frac{1}{4}(v-u)^2$$

$$\square \sim \frac{4}{(v-u)^2} + \frac{1}{4}(-\partial_u + \partial_v) \left[(v-u)^2 (\partial_u + \partial_v) \right]$$

$$+ \frac{4}{(v-u)^2 \sin^2(\phi)} + \frac{4}{\sin^2(v-u)^2}$$

Sec. term:

$$(-\partial_u + \partial_v) \left[(v-u)^2 (-\partial_u + \partial_v) f \right]$$

$$= (-\partial_u + \partial_v) \left[(v-u)^2 (-\partial_u + \partial_v) f \right]$$

$$+ (v-u)^2 \quad r \approx \frac{v}{2}$$

$$\bullet \phi \approx F(u, \theta, \phi) / r$$

$$\sim 2F/v$$

Aside: Introduce retarded coordinates: $u = t - r$

$$ds^2 = -du^2 - 2du dr + r^2 d\Omega^2$$

$$\square \phi = 0$$

$$(-2\partial_u \partial_r + \partial_r^2 + \frac{2}{r}(\partial_r \partial_u) + \frac{1}{r^2} \nabla_{S^2}^2(\theta, \phi)) \phi = 0$$

Assume

$$\phi = r^{-\alpha} \psi(u, \theta, \phi)$$

$$\square \phi = 0 \Rightarrow$$

$$\Rightarrow +2\alpha r^{-\alpha-1} \partial_u \psi + \alpha(1+\alpha) r^{-\alpha-2} \psi - \frac{2}{r} (r^{-\alpha-1} \psi + r^{-\alpha} \partial_u \psi)$$

$$+ \frac{1}{r^2} \nabla_{S^2}^2(\theta, \phi) \phi = 0$$

$$\square \phi \sim 2\alpha r^{-\alpha-1} \partial_u \psi - 2r^{-\alpha-1} \partial_u \psi + \mathcal{O}(r^{-\alpha-2})$$

$$\sim 2(\alpha-1) r^{-\alpha-1} \partial_u \psi$$

$$\bullet \partial_u \psi \neq 0 \text{ i.e. non-zero gradient}$$

→ leading order zero if $\alpha = 1$

$$\phi \sim r^{-1} \psi(r, \theta, \phi) + \mathcal{O}(r^{-2})$$

* Example $\phi(t, r, \theta, \phi) = \phi(t, r)$

$$-\partial_t^2 \phi + \frac{1}{r^2} \partial_r (r^2 \partial_r \phi) = 0$$

$$\phi = r\psi \Rightarrow -\partial_t^2 + \partial_r^2 \psi = 0$$

$$\psi \sim F(t-r) + G(t+r)$$

$$\therefore \phi \sim \frac{1}{r} (F + G)$$

For outgoing: $F(u)$

$$\phi \sim \frac{F(u)}{r}$$

* Introduce $\bar{V} \equiv 1/v$

$$ds^2 = \frac{1}{\bar{V}^2} du d\bar{V} + \frac{1}{4} \left(\frac{1}{\bar{V}} - u \right)^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

• Singular at $\bar{V} = 0$ and so we cannot extend the spacetime metric there.

→ Define $\bar{ds}^2 = \bar{V}^2 ds^2 = \frac{1}{\bar{V}^2} ds^2$

$$\bar{g}_{ab} = \bar{V}^2 g_{ab}$$

Then

$$\bar{ds}^2 = du d\bar{V} + \frac{1}{4} (1 - u\bar{V})^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

which are well behaved at $\bar{V} = 0$

Idea: We expand M^4 by adding to it the pts represented by $\bar{V} = 0$: these are S^2 points with $r = \frac{1}{2}$.

⇒ We can do tensor analysis at " ∞ " (portion of ∞ represented by $\bar{V} \rightarrow \infty$ at constant u).

→ Brought " ∞ " to a finite distance via a conf. trs.

* A better choice: $\tilde{g}_{ab} = \Omega^2 g_{ab}$

$$\Omega^2 = 4(1+v^2)^{-1} (1+u^2)^{-1}$$

$$T \equiv \tan^{-1}(v) + \tan^{-1}(u)$$

$$R \equiv \tan^{-1}(v) - \tan^{-1}(u)$$

⇒

$$T + R = 2 \tan^{-1}(v)$$

$$T - R = 2 \tan^{-1}(u)$$



$\Rightarrow$

$$\begin{aligned} -\pi &\leq T+R \leq \pi \\ -\pi &\leq T-R \leq \pi \end{aligned}$$

$$(R \geq 0)$$

$$R = \tan^{-1}(t+r) - \tan^{-1}(t-r)$$

Suppose $R=0$: $r=0$

• Suppose $R < 0$ i.e. $R = -\alpha, \alpha \in \mathbb{R}^+$

$$-\alpha = \tan^{-1}(t+r) - \tan^{-1}(t-r)$$

$$\alpha = \tan^{-1}(t-r) - \tan^{-1}(t+r) \geq 0$$

(i) $\tan^{-1}(t-r) > \tan^{-1}(t+r)$

$$t-r > t+r$$

or $-r > r$ (a contradiction)

(ii) $\tan^{-1}(t-r) - \tan^{-1}(t+r) > 0$

and $\tan^{-1}(t+r) < 0$

$$t+r < \tan(0) : \underline{t+r < 0}$$

$$\tan^{-1}(t-r) + \alpha \geq 0$$

$$t-r > -\alpha$$

$$t-r < -2r$$

$$\alpha \leq t-r < -2r$$

$$\Rightarrow r < 0 \quad t < \tilde{r}$$

but $r \geq 0$ so this is a contradiction

$\therefore R \geq 0$

* $\underline{\mathcal{L}_S^2 = -dT^2 + dR^2 + \sin^2(R)(d\theta^2 + \sin^2\theta d\phi^2)}$

(Einstein Static universe $S^3 \times \mathbb{R}$)

$\Rightarrow \exists$ a conformal isometry $\psi: M \rightarrow M', (M, g), (M', g')$
 $(\psi^* g)_{ab} = \Omega^2 g'_{ab}$ of Minkowski spacetime
 $(\mathbb{R}^4, \eta_{ab})$ into an open region of the Einstein
 universe $(S^3 \times \mathbb{R}, \tilde{g})$ given by $(R, T, \dots)$

Def: The conformal infinity of M^4 is the boundary $\partial\mathcal{C}$ of $\mathcal{C}$ in the Ein. static universe

- i^- : past timelike infnity ($R=0, T=-\pi$)
- $\mathcal{I}^-$: past null infnity ($T=-\pi+R, 0 < R < \pi$)
- i^0 : Spatial inf. ($R=\pi, T=0$)
- $\mathcal{I}^+$: Future null inf. ($T=\pi-R, 0 < R < \pi$)
- i^+ : Future timelike inf. ($R=0, T=\pi$)

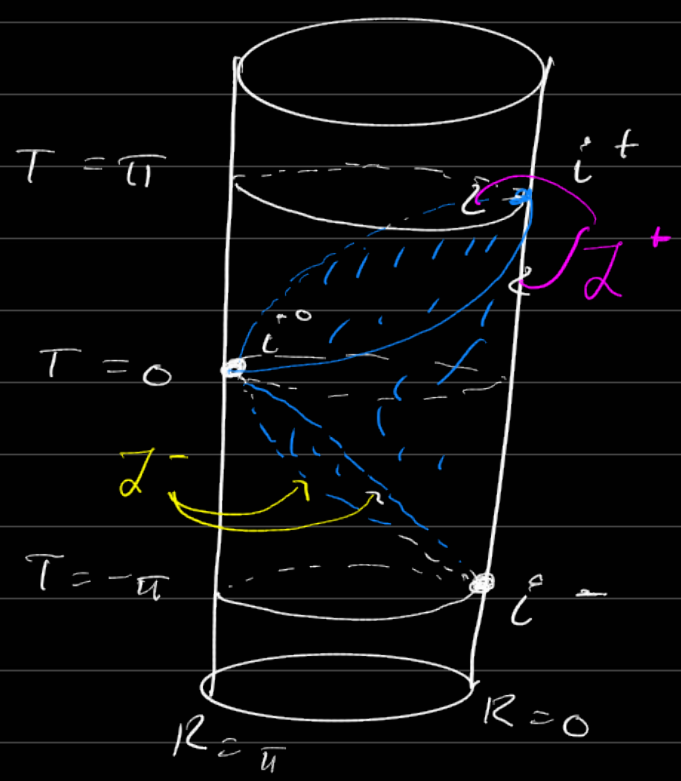
	(T, R)	(u, v)	(t, r)
i^-	$T=-\pi, R=0$	$u \rightarrow \infty, v \rightarrow -\infty$	$t \rightarrow \infty, r \rightarrow -\infty$ $t+r$ fixed
$\mathcal{I}^-$	$T=-\pi+R, 0 < R < \pi$	$u \rightarrow \infty, v$ finite	$r \rightarrow \infty, t \rightarrow -\infty$
i^0	$R=\pi, T=0$	$u \rightarrow \infty, v \rightarrow +\infty$	$r \rightarrow \infty, t$ finite
$\mathcal{I}^+$	$T=\pi-R, 0 < R < \pi$	$v \rightarrow \infty, u$ finite	$r \rightarrow \infty, t \rightarrow \infty$ $t-r$ fixed
i^+	$R=0, T=\pi$	$u \rightarrow \infty, v \rightarrow \infty$	$t \rightarrow \infty, r \rightarrow \infty$

* Note that the space $\mathcal{Q}$ in the Ein. universe:

$$\mathcal{Q} = \mathcal{I}^+(i^-) \cap \mathcal{I}^-(i^+)$$

$$\partial \mathcal{Q} = i^- \cup i^+ \cup \mathcal{I}^- \cup \mathcal{I}^+$$

1c) $\mathcal{Q}$ as $S^1|_{S^2} \times \mathbb{R}$ (suppress an S^2)



or on $(t, r) / (u, v)$:

$$\begin{aligned} -\pi &\leq T+R \leq \pi \\ -\pi &\leq T-R \leq \pi \end{aligned} \quad \Bigg| \quad R \geq 0$$

along every timelike geodesic

λ : affine parameter along a null geo

τ : proper time along a timelike geo.

Proof: $\square \phi = 0$, then $\tilde{\phi} = \Omega^{-1} \phi$

satisfies

$$\tilde{g}^{ab} \tilde{\nabla}_a \tilde{\nabla}_b \tilde{\phi} - \frac{1}{6} \tilde{R} \tilde{\phi} = 0$$

in $\mathcal{O}$

(S)

(sub proof): Conformal Weight: $\psi = \Omega^s \phi$

$$\begin{aligned} \tilde{g}^{ab} \tilde{\nabla}_a \tilde{\nabla}_b (\tilde{\phi}) &= \Omega^{s-2} g^{ab} \nabla_a \nabla_b \phi + (2s + n - 2) \Omega^{s-3} g^{ab} \nabla_a \Omega \nabla_b \phi \\ &\quad + s \Omega^{s-3} \phi g^{ab} \nabla_a \nabla_b \Omega \\ &\quad + s(n-s-3) \Omega^{s-4} \phi g^{ab} \nabla_a \Omega \nabla_b \Omega \end{aligned}$$

$$\rightarrow g^{ab} \nabla_a \nabla_b \phi - \frac{n-2}{4(n-1)} R \phi = 0$$

is conf. invariant.

$$\frac{1}{6} \int R \phi d^4 x$$

Do separate

* On the Cauchy Surface: $T=0$

extend $\tilde{\phi}_0, \tilde{\phi}_0$ to spatial infinity i^0 by setting these to zero there.

$T=0$ still leaves any r so $r \rightarrow \infty$
or $R=\pi$ can be used to set this data.

By thm 10.1.2 $\exists$ a smooth solⁿ $\tilde{\phi}$ throughout the entire $\mathcal{O}$ with the given initial conditions

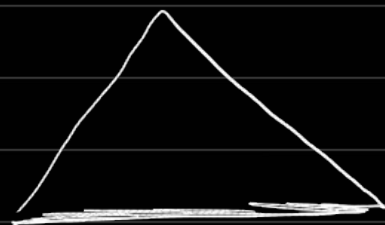
• By uniqueness of the domain $\tilde{\phi}$ agrees with $\tilde{\phi}$ in $\mathcal{O}$.
 $\Rightarrow \tilde{\phi}$ can be smoothly extended to $\mathcal{O}$

• Can extend $\tilde{\phi}$ to $\partial \mathcal{O}$

* From above: $\phi \sim \frac{1}{r}(F+G)$ for Null geodesics
parametrizing $r = r(\lambda)$ $\phi \rightarrow 0$

• Now need similar for timelike: $i^{\pm}$

$$x^\mu(\tau), \quad \phi_\infty = \lim_{\tau \rightarrow \infty} \phi(x(\tau))$$



→ An inertial observer: $x^\mu(\tau) = z^\mu \tau$

$$x^\mu = (\tau, \vec{0})$$

$$\phi(\tau) = \int \frac{d^3k}{(2\pi)^3 2|\mathbf{k}|} (a_{\mathbf{k}} e^{-i|\mathbf{k}|\tau} + a_{\mathbf{k}}^\dagger e^{i|\mathbf{k}|\tau})$$

$$\int k^2 dk d\Omega, \quad k = |\mathbf{k}|$$

$$\phi(\tau) = \frac{1}{(2\pi)^3} \int_0^\infty \frac{k}{2} e^{-i k \tau} \left(\int a(k, \Omega) d\Omega \right) dk + c.c.$$

$$F(k) = \frac{1}{2(2\pi)^2} \int a(k, \Omega) d\Omega$$

$$\phi(\tau) = \int_0^\infty F(k) e^{-i k \tau} dk + c.c.$$

$$\rightarrow \int_0^\infty F(k) e^{-i k \tau} dk = \left[\frac{F(k) e^{-i k \tau}}{-i\tau} \right]_0^\infty + \frac{1}{i\tau} \int_0^\infty F'(k) e^{-i k \tau} dk$$

⇒ N -times

$$I(\tau) = \mathcal{O}(\tau^{-N})$$

$$\downarrow F \xrightarrow{k \rightarrow 0} 0$$

$$F(k) = F(0) + k F'(0) + \dots$$

$$\Rightarrow I(\tau) = \int dk e^{-i k \tau} F(k) = F'(0) \int_0^\infty dk e^{-i k \tau} k$$

$$u'v = uv \Big| - u v'$$

$$u' = e^{-i k \tau} \Rightarrow u = \frac{1}{-i\tau} e^{-i k \tau}$$

$$= \left[\frac{e^{-i k \tau}}{-i\tau} k \right]_0^\infty - \int_0^\infty dk \left(\frac{1}{-i\tau} \right) e^{-i k \tau}$$

$$= \frac{1}{2} F'(0) + \dots$$

$$\therefore \phi(\tau) \sim \mathcal{O}\left(\frac{1}{\tau^2}\right) \quad \tau \rightarrow \infty \quad \text{as claimed}$$

→

Define: A spacetime to be asymptotically flat if the physical spacetime can be mapped isomorphically to a new, "unphysical" spacetime via a conformal isometry with properties similar to M^D

* Properties which do not carry over to curved spacetimes

$$(M, g) \xrightarrow{r \rightarrow \infty} (M^D, \eta)$$

But do not assume that the spacetime

become flat "at a fixed position at early or late times"

- Can't expect cont. wt. to be similar to IM^D at i^- or i^+

(cc) $g_{AB} \rightarrow \eta_{AB}$ as $|r| \rightarrow \infty$

$\Rightarrow$ Smoothness & differentiability of $\tilde{g}$ at i^0 is too strong of a requirement

- (a) All timelike geodesics start at i^- and end at i^+
 - (b) Null geodesics begin at $\mathcal{I}^-$ and end at $\mathcal{I}^+$
 - (c) Spacelike geodesics begin and end at i^0
- $\mathcal{I}^\pm$: chron: timelike
 $\mathcal{I}^\pm$: causal $\sim (\mathcal{I}^\pm \oplus M)$

Def A vacuum spacetime (M, g) is called asymptotically flat at null $(\mathcal{I}^\pm)$ and spatial (i^0) infinity if there exists a spacetime $(\tilde{M}, \tilde{g})$ with $\tilde{g} \in C^\infty$ everywhere except possibly at a point i^0 where it is $C^{2,0}$ and a conformal isometry $\psi: M \rightarrow \psi[M] \subset \tilde{M}$ with conformal factor Ω (so that $\tilde{g} = \Omega^2 \psi^* g$) satisfying: $i^\pm$ are not assumed to exist in $\tilde{M}$

- (i) $\overline{\mathcal{I}^+(i^0)} \cup \overline{\mathcal{I}^-(i^0)} = \tilde{M} - \psi[M]$, i^0 is spacelike related to all points in $\psi[M]$ and the Boundary $\partial\psi[M]$ consists of $\{i^0, \mathcal{I}^+ \equiv \partial\mathcal{I}^+(i^0) \setminus i^0, \mathcal{I}^- \equiv \partial\mathcal{I}^-(i^0) \setminus i^0\}$

(a) topological Boundary is unchanged by taking closures $\partial\mathcal{I}^+(i^0) = \overline{\partial\mathcal{I}^+(i^0)}$

- (ii) There exists an open nbhd, V , of $\partial\psi[M] = i^0 \cup \mathcal{I}^+ \cup \mathcal{I}^-$ s.t. (V, g) is strongly Causal.
- (iii) Ω can be extended to a $f^\pm$ on all of $\tilde{M}$ which is $C^2(i^0)$ and $C^\infty(\text{else})$

(iv) (a) on $\mathcal{I}^+$ and $\mathcal{I}^-$, we have

$$\Omega = 0 \text{ and } \tilde{\nabla}_a \Omega \neq 0 \quad (\mathcal{I}^+, \mathcal{I}^-)$$

$$(b) \quad \Omega(i^0) = 0$$

$$\lim_{i^0} (\tilde{\nabla}_a \Omega) = 0$$

$$\lim_{i^0} (\tilde{\nabla}_a \tilde{\nabla}_b \Omega) = 2 \tilde{g}_{ab}(i^0)$$

* $\tilde{g}_{ab}$ may not be C^1 at i^0 so the limit is taken.

(v) (a) The map of null directions at i^0 into the space of integral curves of $n^a \equiv g^{ab} \tilde{\nabla}_b \Omega$ on $\mathcal{I}^+$ and $\mathcal{I}^-$ is a diffeomorphism.

(b) For a smooth function, w , on $\mathcal{M} - i^0$, $w > 0$ on $\mathcal{M} \cup \mathcal{I}^+ \cup \mathcal{I}^-$ which satisfies

$$\tilde{\nabla}_a (w^4 n^a) = 0 \quad \text{on } \mathcal{I}^+ \cup \mathcal{I}^-, \quad w^{-1} n^a \text{ is complete on } \mathcal{I}^+ \cup \mathcal{I}^-$$

* $C^{2,0}$ condition imposed on $\tilde{g}_{ab}$ at i^0 .

and As one goes along curves in the physical spacetime which $r \rightarrow \pm \infty$ along different (ϕ, θ) then physically the curves become further & further apart. However in the unphysical spacetime the different curves all terminate at i^0 .

• When we consider physical fields on M^D we encounter the direction dependence at i^0

ex: Coulomb sol^n to Maxwell's equation, ΩF_{ab}

approaches a well defined, finite limit as one approaches i^0 along any spacelike geodesic, but the limit defined at i^0 depends on the choice of γ .

$$\rightarrow \lim_{\gamma \rightarrow i^0} (\Omega F) = \Omega \text{sech}^2(\chi) d\tau \wedge (\hat{n}_i dx^i)|_{i^0}$$

$$\text{for spacelike geo: } v^\mu = (\sinh(\chi), \cosh(\chi) \hat{n}^i)$$

$\hat{n}^i$: spatial direction

γ : rapidity parameter

Proof: $ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2$

$Q \equiv q/4\pi$

• For a positive charge at the origin.

$A = -\frac{Q}{r} dt$

thus

$F = dA = -\partial_\mu \left(\frac{Q}{r} \right) dx^\mu \wedge dt$

$= \frac{Q}{r^2} dr \wedge dt = -\frac{Q}{r^2} dt \wedge dr$

$\Rightarrow F = -\frac{Q}{r^2} dt \wedge dr$

$F_{tr} = -Q/r^2 = -F_{rt}, F = 0 \text{ } \forall$

$E_i = F_{i0} = Q/r^2$

$\hookrightarrow$ Electric field $\vec{E} = \frac{Q}{r^2} \hat{r}$

Points radially outward

* Conformal inversion near i^0 :

$x^2 = r^2 - t^2 \geq 0$

• Conformal coords: $y^\mu = \frac{x^\mu}{x^2} = \mathcal{I}(x)^\mu$

$y^\mu = (t, y^i)$

$y^2 = p^2 - t^2$ | $p^2 = \delta_{ij} y^i y^j$

$\hookrightarrow$ Note $x^\mu = \frac{y^\mu}{y^2}$

$t = \tau/y^2, x^i = y^i/y^2$

$y^2 = \frac{1}{x^2} = p^2 - t^2 = \frac{1}{r^2 - t^2}$

$r = p/y^2 \Rightarrow r^2 = \frac{p^2}{y^4} \Rightarrow \frac{r^2}{x^2} = \frac{p^2}{y^2}$

$r^2 \left(\frac{y^2}{x^2} \right) = p^2 = y^2 + t^2$

$y^2 \left(\frac{r^2}{x^2} - \frac{x^2}{x^2} \right) = t^2$

$y^2 x^{-2} (t + t^2) = t^2 = \frac{t^2}{\frac{1}{x^2}}$

$y^2 x^2 = 1 \Rightarrow y^2 = \frac{1}{x^2}$ ✓

Spaced infinity: $r \rightarrow \infty, |t| < r$

$r \rightarrow \infty$

$\frac{p}{p^2 - t^2}$

$\sim \frac{1}{p(1 - \frac{t^2}{p})} \rightarrow \infty$

$p(1 - \frac{t^2}{p}) \rightarrow 0$

$p = 0$ when

• cannot be the case as $r \rightarrow \infty$

$$\text{so } p^2 = \tau^2 \Rightarrow y^2 = 0 \Rightarrow y^m \rightarrow 0$$

since $y^2 > 0$ (spacelike)

$$\Rightarrow r \rightarrow \infty, |b| < r \Rightarrow y^m \rightarrow 0$$

$$i^0: y^m = 0$$

$$ds^2 = -dt^2 + \sum_i dx_i^2$$

$$dt = \frac{d\tau}{y^2} - \frac{2\tau}{y^4} y_m dy^m$$

← giving $\sigma \equiv y^2$

$$\begin{cases} d\tau = \frac{d\tau}{\sigma} \\ dx^i = \frac{x^i}{\sigma} \end{cases}$$

$$-dt^2 = -\frac{d\tau^2}{\sigma^2}$$

$$dx^i \delta_{ij} dx^j = \sigma$$

$$\Rightarrow g_{mn} dx^m dx^n = \sigma^{-2} (-d\tau^2 + \delta_{ij} dy^i dy^j)$$

$$\ast \quad \tilde{g} = \Omega^2 g = -d\tau^2 + \delta_{ij} dy^i dy^j$$

$$\text{at } i^0: y^m \rightarrow 0, \tau \rightarrow 0, y^i \rightarrow 0$$

$$\text{so } \Omega = \sigma = 0$$

$$\begin{aligned} \tilde{\nabla}_a \Omega &= \tilde{\nabla}_a y^2 = \partial_a y^2 = 2 y_m \partial_a y^m \\ &= 2 y_m \delta_a^m = 2 y_a \end{aligned}$$

so

$$\tilde{\nabla}_a \Omega|_{i^0} = 0$$

$$\tilde{\nabla}_a \tilde{\nabla}_b \Omega = \nabla_a (2 y_b) = 2 \partial_a y_b - \tilde{\Gamma}_{ab}^c y_c$$

$$= 2 \partial_a y_b \quad (\tilde{\Gamma} = 0): \text{flat.}$$

$$= 2 \tilde{g}_{ad} \partial^d y_b = 2 \tilde{g}_{ad} \delta_b^d = 2 g_{ab}$$

$$\Rightarrow \tilde{\nabla}_a \tilde{\nabla}_b \Omega = 2 \tilde{g}_{ab}|_{i^0}$$

* For the Coulomb fields:

$$\frac{1}{r^2} = \frac{\sigma^2}{\rho^2}$$

$$dt = (\sigma d\tau - \tau d\sigma)/\sigma^2$$

$$dr = (\sigma d\rho - \rho d\sigma)/\sigma^2$$

$$F = -\frac{Q}{r^2} dt \wedge dr$$

$$= -\frac{Q \sigma^2}{\sigma^4 \rho^2} (\sigma d\tau - \tau d\sigma) \wedge (\sigma d\rho - \rho d\sigma)$$

$$= \frac{-Q}{\sigma^2 \rho^2} (\sigma^2 d\tau \wedge d\rho - \sigma \rho d\tau \wedge d\sigma - \tau \sigma d\sigma \wedge d\rho)$$

$$F = \frac{-Q}{\rho^2} \left(d\tau \wedge d\rho - \frac{\rho}{\sigma} d\tau \wedge d\sigma - \frac{\tau}{\sigma} d\sigma \wedge d\rho \right)$$

$$\Omega F = \frac{-Q}{\rho^2} (\sigma d\tau \wedge d\rho - \rho d\tau \wedge d\sigma - \tau d\sigma \wedge d\rho)$$

$$\begin{cases} d\sigma = 2\rho d\rho - 2\tau d\tau \\ d\sigma \wedge d\rho = -2\tau d\tau \wedge d\rho \\ d\tau \wedge d\sigma = 2\rho d\tau \wedge d\rho \end{cases}$$

$$\begin{aligned} \Rightarrow \Omega F &= \frac{-Q}{\rho^2} (\sigma d\tau \wedge d\rho - 2\rho^2 d\tau \wedge d\rho + 2\tau^2 d\tau \wedge d\rho) \\ &= \frac{-Q}{\rho^2} (y^2 - 2(\rho^2 - \tau^2)) d\tau \wedge d\rho \\ &= \frac{-Q}{\rho^2} (-y^2) d\tau \wedge d\rho \\ &= \frac{Q}{\rho^2} (\rho^2 - \tau^2) d\tau \wedge d\rho \\ &= Q \left(1 - \frac{\tau^2}{\rho^2} \right) d\tau \wedge d\rho \end{aligned}$$

or

$$\Omega F_{(2)} = Q \left(1 - \frac{\tau^2}{\rho^2} \right) d\tau \wedge d\rho$$

(exact in spacetime)
(inversion patch)

* $\rho(y)$ is direction-dependent at $\rho=0$

$$\rho^2 = y^i y_i \quad | \quad 2\rho d\rho = 2y_i dy^i \quad | \quad \rho = |\vec{y}| = \sqrt{y^i y_i}$$

$$\text{or } d\rho = \frac{y_i}{|\vec{y}|} dy^i = n_i dy^i$$

$$n_i \equiv \frac{y_i}{|\vec{y}|} = \frac{y^i}{\rho}$$

$$\Omega F_{(2)} = Q \left(1 - \frac{\tau^2}{\rho^2} \right) n_i d\tau \wedge dy^i$$

neither $\frac{\tau}{\rho}$ nor $n^i = \frac{y^i}{\rho}$

have a unique value at $y^m=0$

• Geodesics in $\tilde{M}^D$ are lines

$$X^\mu(s) = a^\mu + s v^\mu$$

• Spacelike geodesic has $v_\mu v^\mu = 1$

$$-v_0^2 + v_i v^i = 1$$

$$-sh^2 + ch^2 = 1$$

$$sh(\chi) = v_0$$

$$n^i n_i = 1 \quad \text{so}$$

$$V^\mu = (sh(\chi), ch(\chi) n^i)$$

$$x^2(s) = s^2 + 2s(a^\mu V_\mu) + a^2$$

$$\text{as } s \rightarrow \infty \quad a^\mu + s V^\mu$$

$$y^\mu(s) = x^\mu(s)/x^2(s) = \frac{s^2 + 2s(a^\mu V_\mu) + a^2}{s^2} = \frac{V^\mu}{s} + \frac{a^\mu - 2(a \cdot V)V^\mu}{s^2} + \mathcal{O}(s^{-3})$$

(why?)

$$\tau(s) \sim \frac{sh(\chi)}{s}, \quad y^i(s) \sim \frac{ch(\chi) n^i}{s}$$

$$\tau/p \rightarrow th(\chi) \quad n^i = y^i/p = \frac{ch(\chi) n^i s^2}{ch^2(\chi) s} = \frac{s n^i}{ch(\chi)} \rightarrow \hat{n}^i$$

$$\lim_{s \rightarrow \infty} (\Omega F_{(2)}) = \mathcal{Q} (1 - \tanh^2(\chi)) (d\tau \wedge (\hat{n}_i dy^i)) \Big|_{T_{i=0}^* \tilde{M}} \\ = \mathcal{Q} \operatorname{sech}^2(\chi) d\tau \wedge (\hat{n}_i dy^i) \\ \operatorname{sech}^2(\chi) \in (0, 1]$$

Direction dependence:

$$-v_0^2 + (v^i)^2 = 1$$

• Purely space-like: $\chi = 0$

$$v_0 = sh(\chi) = 0$$

(i) Along y^1 :

$$\Rightarrow \chi = 0$$

$$\hat{n}^i dy^i = \hat{n}^{(1)} dy^1$$

$$\lim_{s \rightarrow \infty} (\Omega F_{(2)}) = \mathcal{Q} d\tau \wedge dy^1$$

(ii) Along y^2 :

$$\lim_{s \rightarrow \infty} (\Omega F_{(2)}) = \mathcal{Q} d\tau \wedge dy^2$$

Both different in $\Lambda^2(T_{i=0}^* \tilde{M})$

• Limit remembers the spacelike direction $T_{i=0}^* \tilde{M}$

$$\beta = \tanh(\chi)$$

$$(i) \beta = 0 : \lim (\Omega F_{(2)}) = \mathcal{Q} d\tau \wedge \hat{n}_i dy^i$$

$$(ii) \beta = 1/2 : \lim (\Omega F_{(2)}) = \frac{3\mathcal{Q}}{4} d\tau \wedge \hat{n}_i dy^i$$

Why there cannot be a continuous transition at $i=0$

Suppose $\exists L_{ab}$ at i^0 s.t.

$$\Omega F_{ab} \rightarrow L_{ab}$$

• Then approaching with $\chi = 0$, $\hat{n}_2 = e_1$
 $L^{(2)} = Q d\tau \wedge dy^2$

• " " " $\chi = 0$, $\hat{n} = e_2$
 $L^{(2)} = Q d\tau \wedge dy^2$

ΩF

(i) A finite limit along every spacelike geodesic
 (ii) a well defined limit once the spacelike direction is fixed

(iii) No direction-independent limit at i^0

(iv) Hence no arbitrary extension to i^0

* As seen above, the behavior of $F^{(2)}$ and hence $E^{(1)}$ & $B^{(1)}$ at i^0 depends on the path taken.

Furthermore, its value "remembers" which path is taken. Hence, since we first showed that $F^{(2)}$ has a definite value at i^0 , no such limit of $F^{(2)}(i^0)$ is well defined.

• Thus, the appropriate condition on $\tilde{g}_{ab}$ is that it can be $C^{2,0}$ at i^0 . @

$C^{2,0}$: C^0 ⊕ $\left\{ \begin{array}{l} 1^{st} \text{ derivatives have direction dependent limits at } i^0 \text{ which are smooth in their angular dependence} \end{array} \right\}$

• More precisely, for @:

Let x^u be a smooth coord system with origin at i^0

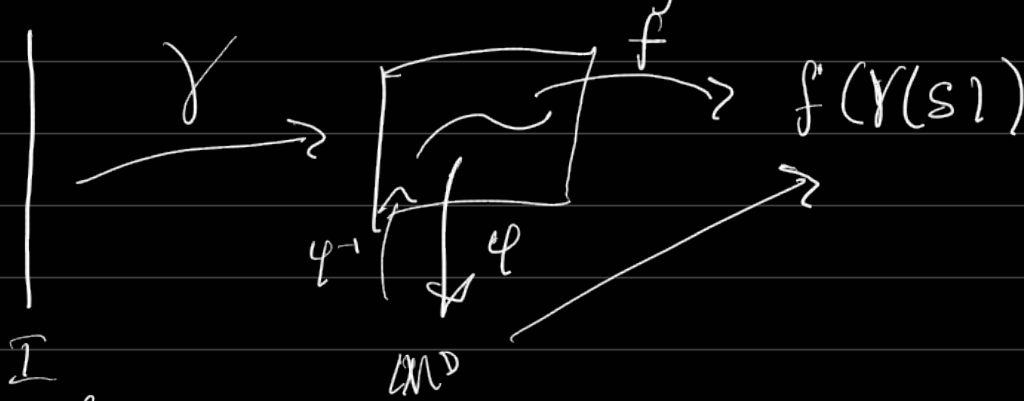
(i) $\rho \equiv \sum_{u=1}^n (x^u)^2$ (Radial f^u)

(ii) define $n-1$ angular functions θ^α ($\alpha=1,2,3$) in terms of x^u (same as 3-sphere coords)

→ use θ^α to characterize each tangent direction at i^0

def: f is said to have a regular direction-dependent limit at i^0 if

(c) For each C^1 curve γ ending at i^0 , the $\lim$ of f along γ exists at i^0



$\lim_{s \rightarrow i^0} f(\phi^{-1}(\phi \circ \gamma)(s))$ exists

• Value of limit depends on the tangent direction to γ at i^0

$$\frac{d\gamma}{d\lambda} = \frac{dx^\mu}{d\lambda} \frac{\partial \gamma}{\partial x^\mu}$$

$$\frac{d}{d\lambda} \sim v^\mu \partial_\mu \quad \lim_{s \rightarrow i^0} = (v^\mu \frac{\partial}{\partial x^\mu}) \gamma$$

• Define $F(\theta^\alpha) = \lim_{s \rightarrow i^0} (f)$

(b) F is smooth on S^3 . $\{\theta^\alpha\}$

(c) Along every C^1 curve ending at i^0 , we have

$$\forall n \geq 1 \quad \lim_{i^0} \frac{\partial^n f}{\partial \theta^n} = \frac{\partial^n F}{\partial \theta^n} \quad \text{and} \quad \lim_{i^0} \left(\rho^n \frac{\partial^n f}{\partial \rho^n} \right) = 0$$

$\Rightarrow$ Precise statement: $\tilde{g}_{ab}$ is $C^{2,0}$:

(i) $\tilde{g}_{ab} \in C^0(\tilde{M})$

(ii) All the first partial derivatives of components of $\tilde{g}_{ab}$ in a smooth chart covering i^0 have regular dependent limits at i^0

• Since p^2 is at the origin (i.e. $p=0$) then the $\lim_{i^0} \left(\rho^n \frac{\partial^n f}{\partial \rho^n} \right)$ does indeed imply that a given f is regular.

• In a chart with $\{\nabla_a x^\mu\}$ orthonormal at i^0 , the components of $\tilde{g}_{ab}$ take the form $\tilde{g}_{\mu\nu} = \eta_{\mu\nu} + \rho \lim(\theta^\alpha) + \mathcal{O}(\rho)$ as

$$\rho \rightarrow 0$$

$\Rightarrow \tilde{g}_{\mu\nu}$ being regular in ρ means that

$$\tilde{g}_{\mu\nu}(p, \theta) \sim \sum p^\alpha \alpha_{\mu\nu}^{(\alpha)}(\theta)$$

$$\tilde{g}_{\mu\nu} = \alpha_{\mu\nu}^{(0)}(\theta) + p \alpha_{\mu\nu}^{(1)}(\theta) + \mathcal{O}(p^2)$$

As we saw above, we found an unphysical metric $\tilde{g} = \Omega^2 g \sim \eta_{\mu\nu}$ so at the limit point $p \rightarrow 0$ (i^0) we expect

$$\tilde{g}_{\mu\nu}(i^0) = \eta_{\mu\nu} = \alpha_{\mu\nu}^{(0)}$$

* Weaker diff structure on $\tilde{M}$ at i^0 to $C^{2,1}$ diff

→ Above is defined for vacuum spacetimes

$$R_{ab} = 0$$

→ Only require $R_{ab} = 0$ "near infinity": MNV

Vinograd of $\mathcal{M}$ where $(V, \tilde{g})$ is strongly causal.

• Can weaken to $\Omega^{-2} T_{ab}$: smooth on $\mathcal{I}^+$ and $\mathcal{I}^-$ has appropriate limiting behavior at i^0 .

$$R_{ab} = \tilde{R}_{ab} + 2\Omega^{-2} \tilde{\nabla}_a \tilde{\nabla}_b \Omega + \tilde{g}_{ab} \tilde{g}^{cd} (\Omega^{-1} \tilde{\nabla}_c \tilde{\nabla}_d \Omega - 3\Omega^{-2} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \Omega)$$

$$= 0$$

↳ Vacuum Einstein field E^{qs} expressed in terms of the unphysical variables

→ Use $\tilde{g}^{cd} \tilde{\nabla}_c \tilde{\nabla}_d \Omega$ over $\tilde{\nabla}^c \tilde{\nabla}_c \Omega$

* High level of arbitrariness in the association of the unphysical spacetime $(\tilde{M}, \tilde{g})$ with (M, g)

→ $\exists$ an equivalence class of unphysical spacetimes all related via conformal factors/symmetries

$$(\tilde{M}, \tilde{g}) \sim (\tilde{M}, \tilde{b}) \Rightarrow \exists \omega \text{ s.t.}$$

$$\tilde{g} = \omega^2 \tilde{b} \quad \text{and} \quad \text{so } g = (\omega \Omega)^2 \tilde{b}$$

provided that $\omega \geq 0$, $C^\infty \setminus C^{1,1}(i^0)$, $C^{2,0}(i^0)$ and $\omega(i^0) = 1$

* Aside: Compare geodesics with respect to $\tilde{\nabla}_a$ and

$$\tilde{\nabla}_a$$

$$\frac{dx^a}{ds}$$

longest vector $V = ds$

$$\frac{d}{ds}(V^a) + \Gamma_{cd}^a V^c V^d = 0$$

$$= V^c \partial_c V^a + \Gamma_{cd}^a V^c V^d$$

$$= V^c (\partial_c V^a + \Gamma_{dc}^a V^d)$$

$$= V^c \tilde{\nabla}_c V^a = 0$$

• $V^a \tilde{\nabla}_a V^b = V^a \nabla_a V^b + V^a C_{ac}^b V^c$

$$= 2 V^b V^c \nabla_c \ln(\Omega) - (g_{ac} V^a V^c) g^{bd} \nabla_d \ln(\Omega)$$

$\Rightarrow \gamma$ fails to be a geodesic wrt $\tilde{\nabla}$ except for null:

$$V^a \tilde{\nabla}_a V^b = 2 V^b V^c \nabla_c \ln(\Omega) = 0$$

$$d\tilde{\lambda}/d\lambda = c\Omega^2$$

* $R_{ab} = \tilde{R}_{ab} + 2\Omega^{-1} \tilde{\nabla}_a \tilde{\nabla}_b \Omega + \tilde{g}_{ab} \tilde{g}^{cd} (\Omega^{-1} \tilde{\nabla}_c \tilde{\nabla}_d \Omega - 3\Omega^{-2} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \Omega)$

$\Rightarrow$ I have -4 (last term)

Meaning of free endstates

(i)-(iv): $(\bar{M}, \tilde{g})$ possesses some of the basic features of the conformal completion of M^D

(i): States i^0 represents spatial ∞ [$\mathcal{I}^+(i^0)$]

(ii): $\exists (V, \phi_a)$ of $\partial\psi[M] = i^0 \cup \mathcal{I}^+ \cup \mathcal{I}^- \cup (V, g_a)$ s. anal.

$\Rightarrow$ No causal pathologies occur near ∞

(iii) Ω is well behaved near ∞

* Key condition (iv):

$$\Omega|_{\mathcal{I}^+} = 0 \text{ and } \tilde{\nabla}_a \Omega \neq 0 \mid \Omega|_{i^0} = 0, \lim_{i^0} (\tilde{\nabla}_a \Omega) = 0$$

$$\tilde{\nabla}_a \tilde{\nabla}_b \Omega = 2 \tilde{g}_{ab} l(i^0)$$

$\Rightarrow$ implies an "infinite amount of stretching" is involved in going from $\tilde{g}_{ab}$ to g_{ab} and $\mathcal{I}^+, \mathcal{I}^-; i^0$ represent " ∞ "

• Derivative conditions imply the physical metric g_{ab} becomes flat as $\rightarrow \infty$

$$\tilde{R} = \Omega^{-2} \{ R - 6 \nabla^a (\Omega^{-1} \nabla_a \Omega) - (2)(3) \Omega^{-2} \nabla_a \Omega \nabla^a \Omega \}$$

$$= \Omega^{-2} \{ R - 6 (1-2) \Omega^{-2} \nabla^a \Omega \nabla_a \Omega + \Omega^{-1} \nabla^a \nabla_a \Omega \}$$

$$\begin{aligned}
& -6 \Omega^{-2} \nabla_a \Omega \nabla^a \Omega \} \\
& = \Omega^{-2} \{ R + 6 \Omega^{-2} \nabla^a \Omega \nabla_a \Omega - 6 \Omega^{-1} \nabla^a \nabla_a \Omega \\
& \quad - 6 \Omega^{-2} \nabla_a \Omega \nabla^a \Omega \} \\
& = \Omega^{-2} \{ R + 6 \Omega^{-2} \tilde{g}^{ab} \nabla_a \Omega \nabla_b \Omega - 6 \Omega^{-1} \tilde{g}^{ab} \nabla_a \nabla_b \Omega \} \\
& = \Omega^{-2} \{ R - 6 \Omega^{-1} D \} \\
& (\Omega^2 R) = \{ R - 6 \Omega^{-1} D \} \\
& \Rightarrow \lim_{\Omega \rightarrow 0} (\Omega^2 R) = (R - 6 D / \Omega)
\end{aligned}$$

but at i^0 $\Omega^{-1} \rightarrow \infty$

$$\lim_{\Omega \rightarrow 0} (g_{ab}(x^\mu)) = \eta_{ab} + \rho \eta_{ab}(\theta^\alpha) + \dots$$

* Coordinate form of the condition on asymptotic behavior of the physical metric as one approaches future null infinity $\mathcal{I}^+$.

$R_{ab} = 0$ (Vacuum Einstein equations)

$$\lim_{\Omega \rightarrow 0} (\Omega R_{ab}) = 0$$

$\hookrightarrow$ find that $\Omega^{-1} \tilde{g}^{cd} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \Omega$ must be smooth at $\mathcal{I}^+$

i.e. $\Omega|_{\mathcal{I}^+} = 0$ and so Ω^{-1} is not defined there.

$$\Omega R_{ab} = 0 = \Omega \tilde{R}_{ab} + 2 \tilde{\nabla}_a \tilde{\nabla}_b \Omega + \tilde{g}_{ab} \tilde{g}^{cd} (\tilde{\nabla}_c \tilde{\nabla}_d \Omega - 3 \Omega^{-1} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \Omega)$$

$$0 = \Omega \tilde{R}_{ab} + 2 \tilde{\nabla}_a \tilde{\nabla}_b \Omega + \tilde{g}_{ab} \tilde{g}^{cd} (\tilde{\nabla}_c \tilde{\nabla}_d \Omega - 3 \Omega^{-1} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \Omega)$$

$$0 = \Omega \tilde{R} + 2 \tilde{\nabla}^a \tilde{\nabla}_a \Omega + D \tilde{g}^{cd} (\tilde{\nabla}_c \tilde{\nabla}_d \Omega - 3 \Omega^{-1} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \Omega)$$

$$0 = \Omega \tilde{R} + (D+2) \tilde{\nabla}^a \tilde{\nabla}_a \Omega - D \Omega^{-1} \tilde{g}^{ab} \tilde{\nabla}_a \Omega \tilde{\nabla}_b \Omega$$

$$\lim_{\Omega \rightarrow 0} (\Omega \tilde{R} + (D+2) \tilde{g}^{ab} \tilde{\nabla}_a \tilde{\nabla}_b \Omega - D \Omega^{-1} \tilde{g}^{ab} \tilde{\nabla}_a \Omega \tilde{\nabla}_b \Omega)$$

$\hookrightarrow \Omega \rightarrow 0$

left with $\lim_{\Omega \rightarrow 0} ((D+2) \tilde{g}^{ab} \tilde{\nabla}_a \tilde{\nabla}_b \Omega - D \Omega^{-1} \tilde{g}^{ab} \tilde{\nabla}_a \Omega \tilde{\nabla}_b \Omega)$

$n^a \equiv \tilde{g}^{ab} \tilde{\nabla}_b \Omega$ is null

$$\begin{aligned}
n^a \tilde{g}_{ab} n^b &= \tilde{g}^{ab} \tilde{\nabla}_b \Omega \tilde{g}_{ac} \tilde{g}^{ce} \tilde{\nabla}_e \Omega \\
&= \tilde{g}^{be} \tilde{\nabla}_b \Omega \tilde{\nabla}_e \Omega = 0 ?
\end{aligned}$$

* first remaining term is smooth at $\mathcal{I}^+$ so

Only $\nabla_a \Omega$ $\tilde{g}^{ab} \nabla_a \Omega \nabla_b \Omega$ remains.

Claim (i): n^a is null:

$$n^a n_a = \tilde{g}^{ab} \tilde{\nabla}_a \Omega \tilde{\nabla}_b \Omega$$

and $\Omega^{-1} n^a n_a$ must be smooth on $\mathcal{I}^+$ for $\Omega = 0$ and so to remedy this we must have $n^a n_a = 0$ and can appropriately L'Hosp. the limit

Claim (ii): n^a is normal to the surface $\mathcal{I}^+$

• $\mathcal{I}^+ = \partial \mathcal{J}^+(i^0) \setminus i^0$ is a null surface

• Ω constant at $\mathcal{I}^+$

↳ The normal 1-form $n_a = \tilde{\nabla}_a \Omega$ on Ω

$X^\mu \in T_{i^0} \tilde{M} \Rightarrow X^\mu n_\mu = 0$ since

$$X^\mu \tilde{\nabla}_\mu \Omega = X^\mu \partial_\mu \Omega = 0$$

$\partial_\mu \Omega$ there

$\Omega = \text{const.}$ (level set)

$d\Omega$ is Normal to Ω

Take a curve $\gamma(\lambda)$ in $\mathcal{I}^+$, then on γ

$$\Omega(\gamma(\lambda)) = 0$$

$$\left. \frac{d}{d\lambda} \Omega(\gamma(\lambda)) \right|_{\gamma(\lambda)} = 0$$

$$\Rightarrow \partial_\mu \Omega \frac{d\gamma^\mu}{d\lambda} = X^\mu \partial_\mu \Omega = X^\mu \nabla_\mu \Omega = X^\mu n_\mu = 0$$

and so X^μ : tangent normal to curve γ in $\mathcal{I}^+$

since γ was arbitrary n^μ is normal to $\mathcal{I}^+$

* Can now use "gauge freedom" in Cent. factor Ω to make $\Omega^{-1} \tilde{g}^{ab} \tilde{\nabla}_a \Omega \tilde{\nabla}_b \Omega$ vanish at $\mathcal{I}^+$

$$\Omega \rightarrow \Omega' = \omega \Omega$$

$$\tilde{g}_{ab} \rightarrow \tilde{g}'_{ab} = \omega^2 \tilde{g}_{ab}$$

$$(\Omega')^{-1} (\tilde{g}')^{cd} \tilde{\nabla}'_c \Omega' \tilde{\nabla}'_d \Omega'$$

$$(\omega \Omega)^{-1} \omega^{-2} \tilde{g}^{cd} \tilde{\nabla}_c (\omega \Omega) \tilde{\nabla}_d (\omega \Omega)$$

$$\omega^{-3} \Omega^{-1} \tilde{g}^{cd} [\omega_c \Omega + \omega \Omega_c] [\omega_d \Omega + \omega \Omega_d]$$

$$= \omega^{-3} \Omega^{-1} \tilde{g}^{cd} [\omega_c \omega_d \Omega + \omega \omega_c \Omega_d + \omega \omega_d \Omega_c + \omega \omega_c \omega_d \Omega_c \Omega_d]$$

$$= \tilde{g}^{cd} \omega^{-3} [\Omega \tilde{\omega}_c \tilde{\omega}_d + 2 \omega \tilde{\omega}_c \tilde{\omega}_d + \Omega^{-1} \tilde{\omega}^2 \tilde{\omega}_c \tilde{\omega}_d]$$

* Normal vector $n^a = \tilde{g}^{ab} \tilde{\nabla}_b \Omega$ on $\mathcal{I}^+$

want

$$(\Omega')^{-1} (\tilde{g}')^{cd} \tilde{\nabla}'_c \Omega' \tilde{\nabla}'_d \Omega' = 0$$

or choose ω to satisfy

$$\boxed{n^a \tilde{\nabla}_a \ln(\omega) = \omega^{-1} (\tilde{g}^{ab} \tilde{\nabla}_a \Omega) (\tilde{\nabla}_b \omega)} \quad \text{on } \mathcal{I}^+$$

$$= -\frac{1}{2} \Omega^{-1} \tilde{g}^{cd} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \Omega$$

$$\Rightarrow \omega^{-3} \tilde{g}^{ab} \Omega \tilde{\nabla}_a \omega \tilde{\nabla}_b \omega + 2 \omega^{-2} \tilde{g}^{cd} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \omega$$

$$- 2 \omega^{-2} \tilde{g}^{ab} \tilde{\nabla}_b \Omega \omega^{-1} \tilde{\nabla}_a \omega$$

$$= \omega^{-3} \tilde{g}^{ab} \Omega \tilde{\nabla}_a \omega \tilde{\nabla}_b \omega$$

which $\rightarrow 0$ on $\mathcal{I}^+$ since $\Omega \rightarrow 0$

• Without loss of generality we can choose the const. factor and $\tilde{g}$ s.t.

$$\Omega^{-1} \tilde{g}^{cd} \tilde{\nabla}_c \Omega \tilde{\nabla}_d \Omega = 0 \quad \text{on } \mathcal{I}^+$$

* This leaves the vacuum Einstein field equations

$$2 \tilde{\nabla}_a \tilde{\nabla}_b \Omega + \tilde{g}_{ab} \tilde{g}^{cd} \tilde{\nabla}_c \tilde{\nabla}_d \Omega = 0 \quad \text{on } \mathcal{I}^+$$

$$2 \delta_a^c \delta_b^d \tilde{\nabla}_c \tilde{\nabla}_d \Omega + \tilde{g}_{ab} \tilde{g}^{cd} \tilde{\nabla}_c \tilde{\nabla}_d \Omega = 0$$

$$(2 \delta_a^c \delta_b^d + \tilde{g}_{ab} \tilde{g}^{cd}) \tilde{\nabla}_c \tilde{\nabla}_d \Omega = 0$$

contract:

$$2 \tilde{\nabla}^2 \Omega + D \tilde{\nabla}^2 \Omega = 0 \Rightarrow (2+D) \tilde{\nabla}^2 \Omega = 0$$

$$(2+D) \geq 0 \Rightarrow \boxed{\tilde{\nabla}^a \tilde{\nabla}_a \Omega = 0} \quad \text{on } \mathcal{I}^+$$

• This implies that null tangent vectors $n^a = \tilde{g}^{ab} \tilde{\nabla}_b \Omega$ on $\mathcal{I}^+$ satisfy the aff. parametric geodesic eq. on $\mathcal{I}^+$ wrt. the unphysical metric

$$n^a \tilde{\nabla}_a n^b = 0$$

$$\hookrightarrow n^a \tilde{\nabla}_a \tilde{g}^{bd} \tilde{\nabla}_d \Omega = n^a \tilde{g}^{bd} \tilde{\nabla}_a \tilde{\nabla}_d \Omega = 0 \quad \checkmark$$

$$\frac{d^2 x^\mu}{ds^2} = \frac{d}{ds} (V^\mu) = V^\sigma \tilde{\nabla}_\sigma V^\mu = 0$$

where $V^\mu = \frac{dx^\mu}{d\tau}$ is aff. parametrized. Here

$$n^a = dy^\mu/d\tau \text{ on } \mathcal{I}^+$$

* Since $B_{ab} = \nabla_a n_b = 0$ on $\mathcal{I}^+$ the expansion, shear and twist of null geodesic generators of $\mathcal{I}^+$ all vanish

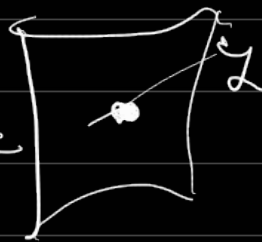
- The gauge charge $n^a \tilde{\nabla}_a h(\omega) = -\frac{1}{2} \Omega^{-1} \tilde{\nabla}^c \epsilon \tilde{\nabla}_c \Omega \tilde{\nabla}_a x$ still permits the additional freedom of choosing ω arbitrarily on any given cross section of $\mathcal{I}^+$ i.e. on a 2D surface, $\mathcal{Z} \subset \mathcal{I}^+$ which intersects each null geodesic generator of $\mathcal{I}^+$ at precisely one point.

$\mathcal{Z} : 2D$

$\Rightarrow \mathcal{Z}$ must be a 2-sphere

and so

$$\mathcal{I}^+ \cong S^2 \times \mathbb{R}$$



$\mathcal{Z}$ intersects at one point.

I don't see the argument here but from a Penrose diagram / conformal coords this is easily seen.

- $\tilde{g}_{ab}$ induces a Riemannian metric $\tilde{h}_{ab}$ on $\mathcal{Z}$

Chp 3, Problem 2: $\nabla_a \nabla^a \alpha = 0 \mid \epsilon_{ab} \epsilon^{ab} = -2$
 $\nabla_a \beta = \epsilon_{ab} \nabla^b \alpha$

(a) Note $\nabla_a \beta$ can be seen as a normal 1-form
 $d\beta = \nabla_a \beta dx^a$, locally $\exists$ a solution β .

β : Harnack conjugate:

$$\begin{aligned} \nabla_a (\nabla^a \beta) &= \nabla^a (\epsilon_{ab} \nabla^b \alpha) \\ &= \frac{1}{2} \epsilon_{ab} \nabla^a \nabla^b \alpha + \frac{1}{2} \epsilon_{ba} \nabla^b \nabla^a \alpha \\ &= \frac{1}{2} \epsilon_{ab} [\nabla^a, \nabla^b] \alpha \\ &= \frac{1}{2} \epsilon^{ab} [\nabla_a, \nabla_b] \alpha \sim \left(\frac{1}{2} R_{ab} \right) dx^a dx^b \\ &= 0 \end{aligned}$$

(b) on M^2 let (α, β) then

$$\nabla^2 \alpha + \nabla^2 \beta = 0 \quad \nabla^2 \alpha = \nabla^2 \beta$$

$$ds^2 = -2d\alpha d\beta \quad (d\alpha^2 + (-1)d\beta^2)$$

$$\alpha = \alpha(x, \tau)$$

$$\beta = \beta(x, \tau)$$

$$ds^2 = -d\tau^2 + dx^2$$

$$d\tau = \frac{\partial \tau}{\partial \alpha} d\alpha + \frac{\partial \tau}{\partial \beta} d\beta$$

$$dx = \frac{\partial x}{\partial \alpha} d\alpha + \frac{\partial x}{\partial \beta} d\beta$$

$$= \left(\left(\frac{\partial \tau}{\partial \alpha} \right)^2 d\alpha^2 + 2 \frac{\partial \tau}{\partial \alpha} \frac{\partial \tau}{\partial \beta} d\alpha d\beta + \left(\frac{\partial \tau}{\partial \beta} \right)^2 d\beta^2 \right)$$

$$+ \left(\left(\frac{\partial x}{\partial \alpha} \right)^2 d\alpha^2 + 2 \frac{\partial x}{\partial \alpha} \frac{\partial x}{\partial \beta} d\alpha d\beta + \left(\frac{\partial x}{\partial \beta} \right)^2 d\beta^2 \right)$$

$$= \left(-\left(\frac{\partial \tau}{\partial \alpha} \right)^2 + \left(\frac{\partial x}{\partial \alpha} \right)^2 \right) d\alpha^2 + \dots$$

$\Rightarrow$ Every Riemann metric on a 2D space is equal to a const. factor times the natural S^2 metric
 $\tilde{h} = f^2 h, \quad (x, u) \cong S^2 \subset \mathbb{R}^3$

* Can now introduce coordinates on a neighborhood of $\mathcal{I}^+$ in the unphysical spacetime $\tilde{M}$

(i) Since $\nabla_a \Omega \neq 0$ on $\mathcal{I}^+$ we may use Ω as a coordinate (since we are not taking $\frac{d\Omega}{d\lambda}$)

(ii) Introduce spherical coords (θ, ϕ) on $\mathcal{I}$ and "carry" these coordinates to other points in $\mathcal{I}^+$ along null geodesic generators of $\mathcal{I}^+$

(iii) r on $\mathcal{I}^+$ to be an affinely parametrized coord along each null geodesic measured from $\mathcal{I}$ with r scaled so that

$$\boxed{R^a \nabla_a r = 1}$$

$$\hookrightarrow \tilde{g}^{ab} \tilde{\nabla}_b \Omega \nabla_a r = 1$$

(iv) extend (r, θ, ϕ) off of $\mathcal{I}^+$ by holding their values fixed along each null geodesic of the family orthogonal to the $\mathcal{I}$ -sphere of constant r of $\mathcal{I}^+$

$$v = t + r$$

* Start with $u = t - r$

$$ds^2 = -du dv + \frac{1}{4}(v-u)^2(d\theta^2 + \sin^2\theta d\phi^2)$$

• Take $V^2 = \frac{1}{v^2}$, then

$$ds^2 = du dv + \frac{1}{u}(1-uv)^2(d\theta^2 + \sin^2\theta d\phi^2)$$

and now let $V = \Omega$

$$d\tilde{s}^2 = du d\Omega + \frac{1}{u}(1-u\Omega)^2(d\theta^2 + \sin^2\theta d\phi^2)$$

As stated in Wald, we can use the residual gauge symmetry to set

$$\left[\begin{array}{l} \frac{1}{u}(1-u\Omega)^2 = 1 \\ \text{or } 1-u\Omega = 2 \Rightarrow u = -\Omega^{-1} ? \end{array} \right]$$

most likely comes from $u\Omega$

$$\left. \frac{d\tilde{s}^2}{du} \right|_u = 2d\Omega du + d\theta^2 + \sin^2\theta d\phi^2$$

• Gauge Condition $\tilde{\nabla}_a \tilde{\nabla}_b \Omega = 0$ on $\mathcal{I}^+$
implies that $\tilde{g}_{\mu\mu}$, $\tilde{g}_{\mu\theta}$ and $\tilde{g}_{\mu\phi}$ are all $\mathcal{O}(\Omega^2)$
as $\Omega \rightarrow 0$

$$\begin{aligned} \tilde{\nabla}_a (\tilde{\nabla}_b \Omega) &= \tilde{\nabla}_a (\partial_b \Omega) \\ &= \partial_a \partial_b \Omega - \tilde{\Gamma}_{ab}^c \partial_c \Omega = 0 \end{aligned}$$

Note

$$\tilde{\nabla}_a \omega_b = \nabla_a \omega_b - C_{ab}^c \omega_c$$

$$C_{ab}^c = \frac{1}{2} \tilde{g}^{cd} \{ \tilde{\nabla}_a \tilde{g}_{bd} + \tilde{\nabla}_b \tilde{g}_{ad} - \tilde{\nabla}_d \tilde{g}_{ab} \}$$

$$= 2\delta_{ca} \tilde{\nabla}_b \ln(\Omega) - g_{ab} \tilde{g}^{cd} \tilde{\nabla}_d \ln(\Omega)$$

→ Show $\partial_a \partial_b \Omega$ for $\forall a, b \neq \Omega$ be

Zero?

→ left with

$$\tilde{\Gamma}_{ab}^c \partial_c \Omega = 0$$

Require $c = \Omega$

$$\tilde{\Gamma}_{ab}^{\Omega} = 0$$

$$\tilde{\Gamma}_{ab}^{\Omega} = \frac{1}{2} \tilde{g}^{\Omega d} (\partial_a \tilde{g}_{bd} + \partial_b \tilde{g}_{ad} - \partial_d \tilde{g}_{ab})$$

$$\bullet \quad (\partial_a \tilde{g}_{bd} + \partial_b \tilde{g}_{ad} - \partial_d \tilde{g}_{ab}) = 0$$

$$\partial_a \tilde{g}_{bu} + \partial_b \tilde{g}_{au} - \partial_u \tilde{g}_{ab} = 0$$

$$\bullet \quad \partial_\Omega \tilde{g}_{uu} + \partial_u \tilde{g}_{\Omega u} - \partial_u \tilde{g}_{\Omega u} = 0$$

$$\partial_\Omega \tilde{g}_{uu} = 0 \Rightarrow$$

$$\text{Thus } \tilde{g}_{uu}|_{\mathcal{I}^+} = 0 \quad \text{and} \quad \partial_\Omega \tilde{g}_{uu}|_{\mathcal{I}^+} = 0$$

Smoothness of $\tilde{g}$ implies: $(\tilde{g} \in C^\infty \text{ away from } \mathcal{I}^+)$

$$g_{uu}(\Omega, u, \theta, \phi) = g_{uu}(0, u, \theta, \phi) + \Omega \partial_\Omega g_{uu}|_{\Omega=0} + \frac{\Omega^2}{2} \partial_\Omega^2 g_{uu}|_{\Omega=0} + \dots$$

or that

$$g_{uu} \sim \mathcal{O}(\Omega^2)$$

$\bullet \quad \tilde{g}_{uu}|_{\mathcal{I}^+} = 0$ from null generator coordinate

* For $\tilde{g}_{u\theta}$

$$\partial_a g_{bc} + \partial_b g_{ac} - \partial_c g_{ab} = 0$$

$$\partial_\Omega (g_{u\theta}) + \partial_u \tilde{g}_{\Omega\theta} - \partial_\theta g_{u\Omega} = 0$$

$$g_{u\Omega} = 1 \Rightarrow \partial_\theta$$

$$\partial_\Omega (g_{u\theta}) = 0 \quad \& \quad g_{u\theta}|_{\mathcal{I}^+} = 0$$

$$\text{So } \tilde{g}_{u\theta} \sim \mathcal{O}(\Omega^2)$$

* For $\tilde{g}_{u\phi}$

$$-\partial_\Omega g_{u\phi} + \partial_u g_{\Omega\phi} + \partial_\phi g_{\Omega u} \rightarrow 0$$

$$\partial_u g_{\Omega\phi} = 0 \Rightarrow$$

$$\tilde{g}_{u\phi} \sim \mathcal{O}(\Omega^2)$$

$\bullet$ In a neighborhood of $\mathcal{I}^+$ $g_{ab} = \Omega^{-2} \tilde{g}_{ab}$

$$\left\{ \begin{aligned} ds^2 &= 2\Omega^{-2} d\Omega du + \Omega^{-2} (ds^2)^2 \\ &+ \text{terms } \mathcal{O}(1) \text{ in } du^2, du d\theta, du d\phi \\ &+ \text{terms } \mathcal{O}(\Omega^{-1}) \text{ in } d\theta^2, d\theta d\phi, d\phi^2, d\Omega du, d\Omega^2 \\ &d\Omega d\theta, d\Omega d\phi. \end{aligned} \right.$$

Take $v = 2/\Omega$ so that $\Omega \rightarrow 0 \Rightarrow v \rightarrow \infty$

$$ds^2 = -dv du + \frac{1}{v} v^2 (d\theta^2 + \sin^2(\theta) d\phi^2)$$

take $v \rightarrow v + f(u, \theta, \phi)$

$$t = \frac{1}{2}(v + u), \quad x = \frac{1}{2}(v - u) \sin(\theta) \cos(\phi)$$

$$y = \frac{1}{2}(v - u) \sinh(\sigma) \sinh(\phi)$$

$$z = \frac{1}{2}(v - u) \cosh(\sigma)$$

$$\Rightarrow ds^2 \sim \frac{1}{r} (-1, 1, 1, 1)$$

so the physical spacetime becomes asymp. M^D

* An initial data set (Σ, h_{ab}, K_{ab})

Σ : 3D-mfd

h_{ab} : Riemannian metric

K_{ab} : symmetric tensor

$\rightarrow$ initial data set

is called asymptotically flat if $\exists$ a mfd $\tilde{\Sigma}$ s.t.

(1) $\Lambda \in \tilde{\Sigma}$ (point at infinity) s.t. $\exists$ a conformal isometry

$$\psi: \Sigma \rightarrow (\tilde{\Sigma} \setminus \Lambda)$$

with

$$\psi^* \tilde{h}_{ab} = \Omega^2 h_{ab} \text{ on } (\tilde{\Sigma} \setminus \Lambda)$$

$$(2) \Omega \in C^{(2)}(\Lambda), \Omega \in C^\infty(\tilde{\Sigma} \setminus \Lambda)$$

$$\tilde{h}_{ab} \in C^{(2)}(\Lambda), \tilde{h}_{ab} \in C^\infty(\tilde{\Sigma} \setminus \Lambda)$$

(3) Ω satisfies conditions at Λ :

$$\bullet \Omega(\Lambda) = 0 \text{ and } \lim_{\Lambda} \tilde{\nabla}_a \Omega = 0$$

$$\bullet \Omega^{-1}(\tilde{\nabla}_a \tilde{\nabla}_b \Omega - 2\tilde{h}_{ab})|_{\Lambda} \rightarrow \text{finite}$$

(4) Unphysical Ricci ${}^{(3)}\tilde{R}_{ab}$ and $\psi^* K_{ab}$ have appropriate limiting behavior at Λ :

$$\Omega^{1/2} {}^{(3)}\tilde{R}_{ab} \text{ and } \Omega \psi^* K_{ab} \rightarrow \text{finite directional dependent limits at } \Lambda.$$

* From an asymptotically flat initial data set one can construct an asymptotically $\mathbb{R}^D$ coord system on the physical initial data surface Σ s.t.

$$\boxed{g_{\mu\nu} \sim \mathcal{O}\left(\frac{1}{r}\right) \text{diag}(-1, 1, 1, 1)} \Big|_{r \rightarrow \infty}$$

and the first derivatives of the components are

$\mathcal{O}(1/r^2)$
 $\rightarrow$ Components of extrinsic curvature. Kays are the
 physical Ricci $R_{ab}^{(3)}$ are,
 $K_{ab} \sim \mathcal{O}(1/r^2)$
 $\sigma R_{ab} \sim \mathcal{O}(1/r^3)$

$$K_{az} \sim \mathcal{O}(\frac{1}{r^2})$$

Problem 2 $\div (\Sigma, u_{ab}, N_{ab})$: Asymptotically flat initial data set

$$S.t. \quad d\bar{X} = \nabla_a \bar{X} (dx^i)^a$$
$$dX = \nabla_a X d(x^a)$$
$$n_{(i)}^a = g^{ab} \nabla_b X_{(i)}$$
$$\begin{aligned} g(\hat{X}, \hat{Y}) &= n_x^a n_y^b g_{ab} \\ &= g^{ac} \nabla_c X \ g^{bd} \nabla_d Y \ g_{ab} \\ &= g^{cd} \nabla_c X \ \nabla_d Y = 0 \\ &= n_y^a \nabla_a X = 0 \\ &= n_x^a \nabla_a Y = 0 \end{aligned}$$

• $\hat{h}_{ab}$ is C^2 at Λ implies

$$R \equiv (x^2 + y^2 + z^2)^{1/2}$$

$$d_0^2 = dx^2 + dy^2 + dz^2$$

$$dX_{\mu} = \nabla_a X_{\mu} (dx^a)^{\mu}$$

$$\hat{\Gamma}_{\alpha\beta}(\sqrt{s}) = \frac{1}{2} \left(\frac{1}{s} + \frac{1}{s'} \right) \left(\frac{1}{s} + \frac{1}{s'} \right) \left(\frac{1}{s} + \frac{1}{s'} \right) \left(\frac{1}{s} + \frac{1}{s'} \right)$$

$$\tilde{h}_{ab}(x) \sim A_{ab} + R \alpha^a \partial_a h_{ab} \quad \text{at } R \rightarrow 0$$

$$\tilde{h}^{ab}(\tilde{\nabla}_a x^i)(\tilde{\nabla}_b x^j) \Big|_{\Lambda} = \delta^{ij}$$

$$\tilde{\nabla}_a x^i = (dx^i)_a$$

• In the limit $R \rightarrow 0$,

$$\tilde{h}^{ab}(R) (dx^i)_a (dx^j)_b = \delta^{ij}$$

$$(\tilde{h}^{ab}(0) + R a^m \partial_m \tilde{h}^{ab} \Big|_{\Lambda}) (dx^i)_a (dx^j)_b = \delta^{ij}$$

$$\lim_{R \rightarrow 0} (\tilde{h}^{ab}(0) \tilde{\nabla}_a x^i \tilde{\nabla}_b x^j) + R a^m \partial_m \tilde{h}^{ab} (\tilde{\nabla}_a x^i)(\tilde{\nabla}_b x^j) = \delta^{ij}$$

$$\boxed{\tilde{h}^{ij}(0) = \tilde{h}^{ab} (dx^i)_a (dx^j)_b = \delta^{ij}}$$

we so

$$\tilde{h}_{ij}(x) = \delta_{ij} + q_{ij}(x)$$

$$\tilde{h}_{ij} \in C^{\geq 0}$$

$$\partial_k \tilde{h}_{ij} \sim \mathcal{O}(1) \text{ in } R \text{ meaning}$$

$$\tilde{h}_{ij}(x) \sim \delta_{ij} + R a^k \partial_k \tilde{h}_{ij}$$

↪ along $t \mapsto tx$

$$\tilde{h}_{ij}(tx) - \tilde{h}_{ij}(0) = \int_0^1 \frac{d}{dt} \tilde{h}_{ij}(tx) dt$$

$$= \int_0^1 \dot{x}^k \partial_k \tilde{h}_{ij}(tx) dt$$

$$x^k = \mathcal{O}(R)$$

$$\therefore q_{ij}(x) \sim \mathcal{O}(R) \Rightarrow q_{ij} = R \alpha_{ij}(\phi, \theta)$$

(2) Claim condition (iii) implies $\Omega = R^2(1 + \mathcal{O}(R))$

$$\Omega(\Lambda) = 0, \quad \lim_{\Lambda} \tilde{\nabla}_a \Omega = 0$$

$$\lim_{\Lambda} \Omega^{-1} (\tilde{\nabla}_a \tilde{\nabla}_b \Omega - 2 \tilde{h}_{ab}) \rightarrow \text{finite}$$

$$\bullet \quad \Lambda \Rightarrow R \rightarrow 0$$

$$\tilde{\nabla}_a \Omega = 0 \quad : \quad \partial_a \Omega|_R = 0$$

$$\Omega(\Lambda) = 0 \Rightarrow \Omega \sim R$$

$$\partial_R \Omega|_{R=0} = 0 \Rightarrow \Omega \sim R^{\geq 2}$$

$$\Omega = R^{\geq 2} f(R, \theta, \phi)$$

$$\lim_{\Lambda} (\tilde{\nabla}_a \tilde{\nabla}_b \Omega - 2 \tilde{h}_{ab}) = 0$$

$$\Rightarrow \tilde{\nabla}_i \tilde{\nabla}_j \Omega = 2 \tilde{h}_{ij}(0)$$

$$\partial_i \partial_j \Omega|_{\Lambda} + \Gamma_{ij}^k (\partial_k \Omega)|_{\Lambda} = 2 \tilde{h}_{ij}(0)$$

$\Rightarrow$ Since $\partial_k \Omega|_{\Lambda} = 0$, we have

$$\begin{aligned} \partial_i \partial_j \Omega|_{\Lambda} &= 2 \delta_{ij} \\ \underline{\Omega(x)}|_{\Lambda} &= \Omega(0)|_{\Lambda} + x^i \partial_i \Omega(0)|_{\Lambda} + \frac{1}{2} x^i x^j \partial_i \partial_j \Omega(0)|_{\Lambda} + \mathcal{O}(R^3) \\ &= \frac{1}{2} (2 \delta_{ij}) x^i x^j + \mathcal{O}(R^3) \\ &= R^2 + \mathcal{O}(R^3) \\ &= R^2 [1 + \mathcal{O}(R)] \end{aligned}$$

• Note: $\partial_k \Omega(x)|_{\Lambda} \sim x^l \partial_k \partial_l \Omega + \mathcal{O}(R^2)$
 $\sim x^l 2 \delta_{kl} + \mathcal{O}(R^2)$

$$\partial_k \Omega(x)|_{\Lambda} \sim 2x_k + \mathcal{O}(R^2)$$

• Define $\tilde{x} \equiv \frac{x}{R^2}$, $\tilde{y} \equiv \frac{y}{R^2}$, $\tilde{z} \equiv \frac{z}{R^2}$

$$r \equiv (x^2 + y^2 + z^2)^{1/2}$$

$$r = (x^2 + y^2 + z^2)^{1/2} \frac{1}{R^2} = \frac{R}{R^2} = \frac{1}{R}$$

$r = \frac{1}{R}$, so a radial conformal rescaling/inversion
 $\Omega \sim \frac{1}{r^2} (1 + \mathcal{O}(r^{-2}))$

$$\tilde{h}_{ij} = \delta_{ij} + \frac{1}{r} \alpha_{ij} \quad (\text{trivially as } r \rightarrow \infty)$$

• Radial unit vector

$$n^i := \frac{x^i}{r} = \frac{x^i}{R}$$

$$\frac{\partial \tilde{x}^k}{\partial x^i} = \frac{\partial}{\partial x^i} \left(\frac{x^k}{r^2} \right) = \frac{1}{r^2} (\delta_i^k - 2n^k n_i)$$

$$= R^2 Q_i^k := R^2 (\delta_i^k - 2n^k n_i)$$

• Q_i^k : Reflection in the plane perpendicular to n^i .

$$Q^T Q = \mathbb{I}$$

$$h_{ij}(x) = \frac{\partial \tilde{x}^k}{\partial x^i} \frac{\partial \tilde{x}^l}{\partial x^j} \tilde{h}_{kl}(\tilde{x})$$

$$= R^4 Q_i^k Q_j^l \Omega^{-2} \tilde{h}_{kl}$$

$$= R^4 Q_i^k Q_j^l \Omega^{-2} (\delta_{kl} + R q_{kl})$$

$$= R^4 Q_i^k Q_j^l \Omega^{-2} (\delta_{kl} + R q_{kl})$$

$$= \alpha_i \alpha_j \frac{2}{r} (1 + \mathcal{O}(n)) \quad (\text{one } + 1 \text{ due})$$

$$\dots \quad h_{ij}(x) \sim \delta_{ij} + \mathcal{O}\left(\frac{1}{r}\right)$$

Claim: First order derivatives of $h_{ij}(x)$ are $\mathcal{O}\left(\frac{1}{r^2}\right)$

as $R \rightarrow \infty$

$$\begin{aligned} \tilde{\partial}_\mu h_{ij}(x) &= \tilde{\partial}_\mu \left(\frac{\partial X^\alpha}{\partial x^i} \frac{\partial X^\beta}{\partial x^j} \tilde{h}_{\alpha\beta} \right) \\ &= \frac{\partial}{\partial x^\mu} \left(\tilde{Q}_i^\alpha \tilde{Q}_j^\beta \tilde{h}_{\alpha\beta} \right) = \frac{\partial}{\partial x^\mu} \left(\frac{1}{r^4} (\delta_i^\alpha - 2n^\alpha n_i) (\delta_j^\beta - 2n^\beta n_j) \right) \\ &= \frac{\partial}{\partial x^\mu} \left(\frac{1}{r^4} (\delta_i^\alpha - 2n^\alpha n_i) (\delta_j^\beta - 2n^\beta n_j) \left(\delta_{\alpha\beta} + \frac{1}{r} \alpha_{\alpha\beta} \right) \right) \\ &= \frac{\partial}{\partial x^\mu} \left(\frac{1}{r^4} \left[\delta_i^\alpha \delta_j^\beta - 2[\delta_i^\alpha n_j^\beta + n_i^\alpha \delta_j^\beta] + 4n_i^\alpha n_j^\beta \right] \right. \\ &\quad \left. \times \left(\delta_{\alpha\beta} + \frac{1}{r} \alpha_{\alpha\beta} \right) \right) \\ &= \frac{\partial}{\partial x^\mu} \left[\frac{1}{r^4} \left(\delta_{ij} - 4[n_i n_j] + 4n_i n_j \right. \right. \\ &\quad \left. \left. + \frac{\alpha_{ij}}{r} - \frac{2}{r} [\alpha_{i\beta} n_j^\beta + \alpha_{\alpha j} n_i^\alpha] \right. \right. \\ &\quad \left. \left. + \frac{1}{r} n^\alpha \alpha_{\alpha\beta} n^\beta n_i n_j \right) \right] \\ &= \partial_\mu \left[\frac{1}{r^4} \left\{ \delta_{ij} + \frac{\alpha_{ij}}{r} - \frac{4}{r} \alpha_{\alpha(i} n_{j)} n^\alpha + \frac{1}{r} n^\alpha \alpha_{\alpha\beta} n^\beta n_i n_j \right\} \right] \\ &\bullet \partial_\mu n^i \sim \mathcal{O}\left(\frac{1}{r}\right) \end{aligned}$$

$\dots$ Find that $\partial_{x^\mu} h_{ij}(x) \sim \mathcal{O}\left(\frac{1}{r^2}\right)$

*Claim: (iv) implies that components of K_{ab} and ${}^{(3)}R_{ab}$:

$$K_{ab}(x) \rightarrow \mathcal{O}\left(\frac{1}{r^2}\right)$$

$${}^{(3)}R_{ab}(x) \rightarrow \mathcal{O}\left(\frac{1}{r^3}\right)$$

(iv) $\Omega^{1/2} {}^{(3)}R_{ab}$ and $\Omega^{3/2} K_{ab} \rightarrow \text{finite}$

•

$$\Omega = R^2 (1 + \mathcal{O}(R))$$

$$\hookrightarrow \tilde{\Omega} = \frac{1}{r^2} (1 + \mathcal{O}\left(\frac{1}{r}\right))$$

• $A_{ab} \equiv \tilde{\Omega} K_{ab}$ is bounded at Δ

$$|A_{ab} - \omega| < \epsilon \quad \forall a, b$$

as $|x - \Lambda| < \delta$

• Since A_{ab} is bounded then

$$A_{ab} \sim \mathcal{O}(r^{\leq 0})$$

$$\leq 0: A_{ab} \sim \mathcal{O}(r^{\leq 0})$$

Then as $r \rightarrow \infty$ $A_{ab} \rightarrow 0$
and so A has compact support and
eventually goes to zero.

→ Since each are direction dependent then
they cannot all go to zero: this
is the same limit.

Thus

$$A_{ab} \sim \mathcal{O}(1)$$

$$\text{and } A_{ab} = \Omega K, \quad \Omega \sim \mathcal{O}(R^2)$$

$$\text{so } K \sim \mathcal{O}(R^{-2})$$

$$K_{ab}(x) = \frac{\partial x^i}{\partial \bar{x}^a} \frac{\partial x^j}{\partial \bar{x}^b} K_{ij}(x) \quad \left(A \sim \mathcal{O}(R^{-2}) \right)$$

$$x^i = \bar{x}^i / R^2$$

$$\Rightarrow \frac{\partial x^i}{\partial \bar{x}^a} = \frac{\delta_a^i}{R^2} - \frac{2 \bar{x}^i \bar{x}_a}{(R^2)^2}$$

$$= \frac{\delta_a^i}{R^2} - 2 \bar{x}^i \bar{x}_a$$

$$= \frac{1}{R^2} (\delta_a^i - 2 n^i n_a)$$

$$= r^2 (\delta_a^i - 2 n^i n_a)$$

∴

$$K_{ij}(x) = \frac{\partial \bar{x}^a}{\partial x^i} \frac{\partial \bar{x}^b}{\partial x^j} K_{ab}(\bar{x})$$

$$x^i = \bar{x}^i / R^2 \Rightarrow \bar{x}^i = \frac{x^i}{r^2}$$

$$\frac{\partial \bar{x}^a}{\partial x^i} = \frac{\delta_i^a}{r^2} - 2 \frac{\bar{x}^a \bar{x}_i}{r^2}$$

$$= \frac{1}{r^2} (\delta_i^a - 2 n^a n_i)$$

$$\begin{aligned}
\mathcal{U}_{ij}(x) &= R^{\mu} Q_i^{\alpha} Q_j^{\beta} \mathcal{U}_{\alpha\beta}(x) \\
&\sim \mathcal{O}(R^4) \mathcal{O}(1) \mathcal{O}(R^{-2}) \\
&\sim \mathcal{O}(R^2) \\
&\sim \mathcal{O}\left(\frac{1}{r^2}\right)
\end{aligned}$$

* Now for

$$\Sigma'^{12}{}^{(3)} R_{ab} \sim \mathcal{O}(1)$$

by the same argument from above.

$$\Sigma'^{12} \sim \mathcal{O}(R)$$

$$R_{ab} \sim \mathcal{O}\left(\frac{1}{R}\right) \text{ in } \{X^i\}$$

$$\begin{aligned}
\therefore \hat{R}_{\alpha\beta}(x) &= \frac{\partial X^{\alpha}}{\partial x^i} \frac{\partial X^{\beta}}{\partial x^j} R_{\alpha\beta}(x) \\
&= R^{\mu} Q_i^{\alpha} Q_j^{\beta} R_{\alpha\beta}
\end{aligned}$$

and so

$$\begin{aligned}
\hat{R}_{ij}(x) &\sim \mathcal{O}(R^4) \mathcal{O}(1) \mathcal{O}\left(\frac{1}{R}\right) \\
&\sim \mathcal{O}(R^3) \\
&\sim \mathcal{O}\left(\frac{1}{r^3}\right)
\end{aligned}$$

as claimed (End of Problem 2)

Question 5:

• States that $\mathcal{I}^+$ and $\mathcal{I}^-$ have the "right size"

(a) Null geodesic generators of $\mathcal{I}^+$ and $\mathcal{I}^-$ emanate from i^0 in an appropriate way.

$$\mathcal{I}^+ \sim_{\text{top}} S^2 \times \mathbb{R}$$

$$\mathcal{I}^- \sim_{\text{top}} S^2 \times \mathbb{R}$$

• Since $\tilde{g}$ is only C^2 at i^0

(b) "all of $\mathcal{I}^+$ and $\mathcal{I}^-$ to be present" in M_c^p

$$\Leftrightarrow \tilde{\nabla}_a \tilde{\nabla}_b \Omega = 0 \text{ on } \mathcal{I}^+$$

(null geodesic generators of $\mathcal{I}^+$ and $\mathcal{I}^-$ are complete)
 $\hookrightarrow \tilde{\nabla}_a (\omega^{\mu} n_a) = 0$

$$\text{is equivalent to } n^a \tilde{\nabla}_a h(\omega) = -\frac{1}{2} \Omega^{-1} \tilde{g}^{ab} \tilde{\nabla}_a \Omega \tilde{\nabla}_b \Omega$$

$$(n^a = \tilde{g}^{ab} \tilde{\nabla}_b \Omega)$$

"Proof":

$$4\pi \int n^a \tilde{\nabla}_a \Omega + 16\pi \tilde{\nabla}_a n^a = 0$$

$$\begin{aligned} & \psi \omega^{-1} n^a \tilde{\nabla}_a \omega + \tilde{\nabla}_a n^a = 0 \\ \Rightarrow & \psi n^a \tilde{\nabla}_a \ln(\omega) + \tilde{\nabla}_a (\tilde{g}^{ab} \tilde{\nabla}_b \omega) = 0 \\ & \psi n^a \tilde{\nabla}_a \ln(\omega) + \tilde{\nabla}_a \tilde{g}^{ab} \tilde{\nabla}_b \omega = 0 \end{aligned}$$

$$\begin{aligned} \tilde{\nabla}_a \tilde{g}_{bc} &= 2 \Omega g_{bc} \nabla_a \Omega \\ \tilde{\nabla}_a \tilde{g}_{bc} &= 2 \Omega g_{bc} \nabla_a \Omega - \Gamma^g_a - \Gamma^g_a \\ &= 2 \Omega^{-1} \tilde{g}_{bc} \nabla_a \Omega - \Gamma^g_a - \Gamma^g_a \\ \Rightarrow & \text{only first term survives} \end{aligned}$$

$$\rightarrow \psi n^a \tilde{\nabla}_a \ln(\omega) + (2 \Omega^{-1} \tilde{g}^{ab} \tilde{\nabla}_a \Omega \tilde{\nabla}_b \Omega) = 0$$

and so

$$n^a \tilde{\nabla}_a \ln(\omega) = -\frac{1}{2} \Omega^{-1} \tilde{g}^{ab} \tilde{\nabla}_a \Omega \tilde{\nabla}_b \Omega$$

as claimed.

* Completeness of null geodesics generators of $\mathcal{I}^+$ is eq. to the statement that in our coord. system (x, u, θ, ϕ)

$$u = (-\infty, \infty)$$

(i) (M, g) is said to be asymptotically flat at spatial
 i^∞ if $\exists$ a spacetime $(\bar{M}, \bar{g})$ with $\bar{g}_{ab} \in C^\infty(\bar{M} - i^0)$
 where $\bar{g}_{ab} \in C^{2,0}(i^0)$, $\exists \psi: M \rightarrow \psi[M] \subset \bar{M}$ with
 conf. factor $\Omega: \{1, 2, 3, 4, 5\}$

(ii) A vacuum spacetime (M, g) : asymptotically flat
 at null ∞ [$\mathcal{I}^+$] $\exists$ a mfd with Boundary, $\bar{M}$, with
 smooth metric $\bar{g}$ and a conformal isometry of M
 onto the interior of $\bar{M}$ with $\Omega \in \{1, 2, 3, 4, 5\}$
 all hold.

$\rightarrow \bar{\mathcal{I}} = \partial \bar{M}$ and in place of $\mathcal{I}$: require $\bar{\mathcal{I}}$

$$\bar{\mathcal{I}} = \bar{\mathcal{I}}^+ \sqcup \bar{\mathcal{I}}^-$$

$$\bar{\mathcal{I}}^+ \cap \bar{\mathcal{I}}^-(\text{int}(M)) = \emptyset \quad \& \quad \bar{\mathcal{I}}^- \cap \bar{\mathcal{I}}^+(\text{int}(M)) = \emptyset$$

* $(\bar{M}, \bar{g})$: Must be strongly causal in a nbhd of

Asymptotic symmetries:

• Minkowski: (M^4, η) has

(i) 4-translations

(ii) $SO(4)$ $O^T O = \mathbb{1}$, $\det(O) = 1$

$$(1^T + i \xi 1^T)(1 + i \xi T) = \mathbb{1}$$

$$\mathbb{1} + i \xi T^T + i \xi T = \mathbb{1}$$

$$\therefore T^T = -T \leftarrow \text{Anti-symmetric}$$

$$T: (n^2 - n)/2 \quad n=4 \quad 1^2/2 = \underline{6}$$

$SO(4) \sim \underline{6}$ Rotations

$$\therefore \underline{4} \oplus \underline{6} = \underline{(10)} \quad (\text{Poincare})$$

10-parameter group of isometries: $Isot^+(1,3)$

* In a general curved spacetime one would not expect any exact isometries to be present.

→ Since asymptotically flat S.T. approaches Minkowski and hence it may be possible to define symmetries at i^0 or $\mathcal{I}^\pm$.

→ These are ∞ -dim subgroup of "angle dependent translations": supertranslations.

• Idea: Want an intrinsic asymptotic symmetry at future null infinity to be represented by an equiv. class of vector fields $[\xi^a]$ in (M, g) s.t.

$$(\mathcal{L}_\xi g)_{ab} = 0$$

"as good an approximation as possible" in the limit $\rightarrow \mathcal{I}^\pm$

(i) ξ^a : viewed as a vector field in $\tilde{M}$ $(\varphi_* \xi^a)$ have a smooth extension to $\mathcal{I}^\pm$

(ii) $\Omega^2 \mathcal{L}_\xi g_{ab}$ also has a smooth extension to $\mathcal{I}^\pm$ s.t. $\lim_{\mathcal{I}^\pm} (\Omega^2 \mathcal{L}_\xi g_{ab}) = 0$

(iii) Two vector fields ξ^a and ξ'^a on the physical S.T. solutions to above conditions if

$\times$ S.T. satisfying the above symmetry

$$\lim_{\mathcal{I}^+} (\psi * \xi^a) = \lim_{\mathcal{I}^+} (\psi * \xi'^a)$$

$\times$ The asymptotic group is universal: All asymptotically flat S.T. generate the same group.

→ **BMS-group** (Bondi-Metzner-Sachs)

MP:

$$v = t + r, \quad u = t - r$$

$$ds^2 = -dr dv + \frac{1}{r^2} (v - u)^2 (d\theta^2 + \sin^2(\theta) d\phi^2)$$

For an arbitrary function $f = f(\theta, \phi)$

$$\xi^a = f \left(\frac{\partial}{\partial u} \right)^a + \frac{v}{2r^2} \frac{\partial f}{\partial \theta} \left(\frac{\partial}{\partial \theta} \right)^a + \frac{v}{2r^2 \sin^2(\theta)} \frac{\partial f}{\partial \phi} \left(\frac{\partial}{\partial \phi} \right)^a$$

is non-vanishing at $\mathcal{I}^+$

But $\Omega^2(\mathcal{L}_\xi g)_{ab}$ vanishes at $\mathcal{I}^+$

$\times$ In a nonflat but Asymptotically flat S.T. one can find any ξ s.t. Killing eq. near $\mathcal{I}^+$ to a better approx. than this.

$\bullet$ f is a f^m on the celestial sphere.

$$\xi^a = B^M (N_M)^a$$

$$(N_M)^a = \left(\frac{\partial^a}{\partial u}, \frac{\partial^a}{\partial v}, \frac{\partial^a}{\partial \theta}, \frac{\partial^a}{\partial \phi} \right)$$

$$B^M = \left(f, 0, \frac{v}{2r} \frac{\partial f}{\partial \theta}, \frac{v}{2r^2 \sin^2(\theta)} \frac{\partial f}{\partial \phi} \right)$$

→ Something of this form?

$$\begin{aligned} (\mathcal{L}_\xi g)_{ab} &= \xi^M \partial_M g_{ab} + \partial_a V^M g_{Mb} + \partial_b V^M g_{aM} \\ &= \partial_a \xi^M h_{Mb} + \partial_b \xi^M h_{aM} \end{aligned}$$

$\bullet \frac{v}{2r^2} \nabla^A f \partial_A$: Compensating diffeomorphism required to preserve Asymp. form

$$v = u + 2r \Rightarrow v/2r^2 = \frac{1}{r} + \frac{u}{2r^2}$$

$$\xi^A = \frac{1}{r} D^A f + \mathcal{O}(r^{-2})$$

$$(u, \theta, \phi) \mapsto (u + f(\theta, \phi), \theta, \phi)$$

Bondi-Coordinates: (u, r, θ, ϕ)

$$ds^2 = -dr^2 - 2dr d\Omega^2 + r^2 d\Omega^2$$

$$\Omega^2 = \frac{1}{r^2}$$

$$ds^2 = \frac{1}{r^2} (-r^2 dr^2 - 2r^2 dr d\Omega^2 + d\Omega^2)$$

$$d\tilde{s}^2 = (-\frac{1}{r^2} dr^2 - 2r^{-2} dr d\Omega^2 + d\Omega^2)$$

$$\rho \equiv \frac{1}{r} = \Omega$$

$$d\rho = -r^{-2} dr$$

$$d\tilde{s}^2 = (-\rho^2 dr^2 + 2 d\rho dr + d\Omega^2)$$

Regular at $r \rightarrow \infty \iff \rho \rightarrow 0$

$$(d\tilde{s}^2)|_{\mathcal{I}^+} \approx 2 d\rho dr + d\Omega^2$$

• Normal component to $\mathcal{I}^+$

$$n_a = \tilde{\nabla}_a \Omega = \partial_a \rho$$

$$n^a = \tilde{g}^{ab} \tilde{\nabla}_b \Omega = g^{ap} \tilde{\nabla}_p \Omega$$

$$= g^{ap}$$

$$\rightarrow n_a d\rho : n_n = \tilde{\nabla}_n \rho = \partial_n \rho$$

$$\tilde{g}^{mn} n_m n_n = \tilde{g}^{pp} = \rho^2 \rightarrow 0$$

$\mathcal{I}^+$ which is a null Hyper surface

$$\text{claim: } n^M \sim \frac{\partial}{\partial u}$$

$$\tilde{g}_{(n,p)} = \begin{pmatrix} \tilde{g}_{uu} & \tilde{g}_{ur} \\ \tilde{g}_{ur} & \tilde{g}_{rr} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & \rho^2 \end{pmatrix}$$

$$n^a = \tilde{g}^{ab} \tilde{\nabla}_b \Omega = \tilde{g}^{ab} \partial_b \rho$$

$$= g^{ap}$$

$$n = n^a \partial_a = n^u \frac{\partial}{\partial u} + \rho^2 \frac{\partial}{\partial \rho}$$

However at $\mathcal{I}^+$ the second term $\rightarrow 0$

$$n|_{\mathcal{I}^+} \sim \frac{\partial}{\partial u}$$

* Tangent to $\mathcal{I}^+$ is spanned by n in the direction $\frac{\partial}{\partial u}$

* First term in ξ^a is in this direction

$$\xi_{(1)} = f \left(\frac{\partial}{\partial u} \right)$$

$\Rightarrow$ Second term is a gradient correction term

$$\frac{1}{2} \nabla_A \rho \partial_A$$

$$\tilde{\nabla}^A f = \partial_t f \partial_t + \csc(\theta)^2 \partial_\phi f \partial_\phi$$

* Asymptotic symmetries of physical l.t. extend to transformations of $\mathcal{I}^+$ in the unphysical.

$$\begin{aligned} (n')^a &= (\tilde{g}^{ab})' \tilde{\nabla}'_b \Omega' \\ &= \omega^{-2} \tilde{g}^{ab} (\tilde{\nabla}'_b \Omega') = \omega^{-2} \tilde{g}^{ab} (\tilde{\nabla}_b (\omega \Omega)) \\ &= (\omega^{-2} \tilde{g}^{ab} \tilde{\nabla}_b \omega) \Omega + \omega^{-2} \tilde{g}^{ab} (\tilde{\nabla}_b \Omega) \omega \\ &= \omega^{-2} \tilde{g}^{ab} (\tilde{\nabla}_b \omega) \Omega + \omega^{-1} \tilde{g}^{ab} \tilde{\nabla}'_b \Omega \\ &= \underbrace{\omega^{-2} \tilde{g}^{ab} (\tilde{\nabla}_b \omega) \Omega}_{\text{term vanishes? or is cancelled by } \tilde{\nabla}' \text{ term}} + \omega^{-1} n^a \end{aligned}$$

$$\hookrightarrow \tilde{\nabla}'_b \Omega = \partial_b \Omega \quad \text{and} \quad \tilde{\nabla}'_a \omega = \partial_a \omega$$

$$\begin{aligned} (n')^a &= \tilde{g}^{ab} \tilde{\nabla}'_b (\Omega') \\ &= \omega^{-2} \tilde{g}^{ab} \partial_b (\omega \Omega) \\ &= \omega^{-2} \tilde{g}^{ab} ((\partial_b \omega) \Omega + \omega (\partial_b \Omega)) \end{aligned}$$

In the limit $\mathcal{I}^+$: $\Omega \rightarrow 0$ and $\nabla_b \Omega \neq 0$

$$(n')^a|_{\mathcal{I}^+} = \omega^{-1} n^a$$

BMS Group: $\psi: \mathcal{I}^+ \xrightarrow{\text{diffeo}} \mathcal{I}^+$ s.t. $\exists \omega, \alpha(\omega)$

$$\psi^* \tilde{h}_{ab} = \omega^2 \tilde{h}_{ab}$$

and $(\psi_* n^a) = \alpha n^a$ s.t. $\alpha = \alpha(\omega)$

$\hookrightarrow \psi$: preserve the universal structure.

* $\tilde{g}_{ab}$ induces a degenerate metric $\tilde{h}_{ab}$ on a null hypersurface $\mathcal{I}^+ \cong S^2 \times \mathbb{R}$, $\forall a, b, \tilde{h}^{ab} = 0$

• n^a : tangent to $\mathcal{I}^+$: vector field on $\mathcal{I}^+$

$$n^a = \tilde{g}^{ab} \tilde{\nabla}_b \Omega$$

• Under $\Omega \rightarrow \Omega' \equiv \omega \Omega$

$$\tilde{h}_{ab} \rightarrow \tilde{h}'_{ab} = \omega^2 \tilde{h}_{ab}$$

$$n^a \rightarrow (n')^a = \omega^{-1} n^a|_{\mathcal{I}^+}$$

$\Rightarrow$ Let $(\mathcal{I}^+, (\tilde{h})_{ab}, (n)^a)$ be the future null ∞ , induced

well define, and induced vector field associated with the physical spacetime (M_1, g_{1ab})

$(\mathcal{I}_2^+, (\tilde{h}_2)_{ab}, (n_2)^a)$: corresponding structure for the asymptotic spacetime $(\tilde{M}, \tilde{g}_{ab})$.

c.c

$$(\mathcal{I}_{(1)}^+, (\tilde{h}_1)_{ab}, (n_1)^a) \hookrightarrow (M_{(1)}, g_{(1)ab}) = (M, g_{ab})$$

$$(\mathcal{I}_{(2)}^+, (\tilde{h}_2)_{ab}, (n_2)^a) \hookrightarrow (M_{(2)}, g_{(2)ab}) = (\tilde{M}, \tilde{g}_{ab})$$

$\Rightarrow$ choices of conformal gauge in the two unphysical spacetimes s.t. there is a diffeomorphism

$$\psi: \mathcal{I}_{(1)}^+ \rightarrow \mathcal{I}_{(2)}^+$$

$$\psi^*(\tilde{h}_1)_{ab} = (\tilde{h}_2)_{ab}$$

$$\psi^*(\tilde{n}_1)^a = (n_2)^a$$

• Proof: For any $\mathcal{I}^+ \ni$ a cert gauge and coords

(u, θ, ϕ) s.t.

$$h_{ab} = (d\theta)_a (d\theta)_b + \sin^2(\theta) d\phi_a d\phi_b$$

$$\text{and } n^a = \left(\frac{\partial}{\partial u}\right)^a$$

Let ψ be the map which associates with each point on $\mathcal{I}_1^+$ the point on $\mathcal{I}_2^+$ with the same value of these coordinates.

Supertranslations: Vector fields on $\mathcal{I}^+$ of the form

$$\xi^a = \alpha n^a$$

α : constant on each generator of $\mathcal{I}^+$

$$\hookrightarrow n^a \tilde{\nabla}_a \alpha = 0$$

$$\text{BMS} \supset \text{Supertranslations} \equiv \mathcal{S}$$

$\hookrightarrow$ " ∞ "-dim normal subgroup.

$$\text{BMS}/\mathcal{S} \cong \text{SO}(1,3)$$

* Ricci tensor vanishes near $\mathcal{I}^+$ & i^0 so the physical Weyl

tensor $C_{abc}{}^d$: contains all the information about the physical curvature.

- $C_{abc}{}^d$ is conformally invariant

$$C_{abc}{}^d = \tilde{C}_{abc}{}^d \text{ on } M$$

Can show that $\tilde{C}_{abc}{}^d$ must vanish at $\mathcal{I}^+$

- Let γ be any null geodesic in $\tilde{M}$ from $p \in M$
 $q \in \mathcal{I}^+$

$$\gamma: M \rightarrow \mathcal{I}^+ \cup M$$

$$[p, \tilde{q}] \mapsto [p, q]$$

γ is a curve starting at $p \in M$ and ending at $q \in \mathcal{I}^+$

- Take λ to be an affine parameter ($\lambda \sim \alpha \tau + \beta$,
s.t. $q = \lim_{\lambda \rightarrow \infty} \gamma(\lambda x)$)

Get an 'inf' amount of "stretching" in the K^a direction

$\Rightarrow C_{abc}{}^d$ follows the "peeling property"

$$C_{abcd} = \frac{C_{abcd}^{(1)}}{\lambda} + \frac{C_{abcd}^{(2)}}{\lambda^2} + \frac{C_{abcd}^{(3)}}{\lambda^3} + \frac{C_{abcd}^{(4)}}{\lambda^4} + \mathcal{O}\left(\frac{1}{\lambda^5}\right)$$

in the limit $\lambda \rightarrow \infty$

$C_{abcd}^{(1)}$ is bounded asymptotically

Energy: T_{ab}

- The total energy of the field associated with a time-translator Killing field t^a on a spacelike Cauchy Hypersurface Σ

$$E = \int_{\Sigma} T_{ab} n^a t^b$$

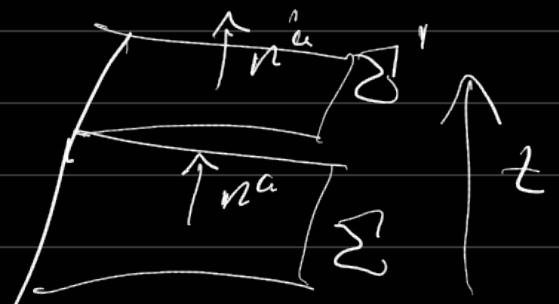
n^a : unit normal to Σ

$$(\partial_t g)_{ab} = 0 = \nabla_a t_b + \nabla_b t_a$$

$$n^a \sim t^a / t^2$$

- $\partial_a T_{ab} = 0$ Guarantees the

$$\text{Note } \int T_{ab} n^a t^b = \int d^3x$$



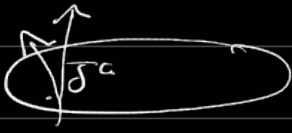
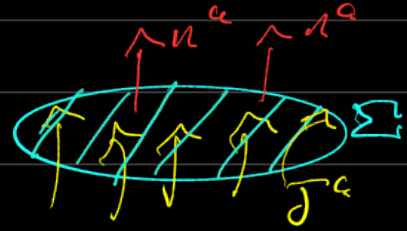
• The current associated to t^a is

$$J^a = T^{ab} t_b \quad | \quad \nabla_a T^{ab} = 0, T^{[ab]} = 0$$

$$\nabla_a J^a = \nabla_a (T^{ab} t_b)$$

$$\Rightarrow \nabla_a J^a = 0$$

Flo~~x~~ of J^a through Σ :



$$\frac{dQ}{dA} = n^a J_a \Rightarrow dQ = n^a J_a dA$$

$$Q = \int_{\Sigma} n^a J_a d\Sigma = \int_{\Sigma} n^a T_{ab} t^b d\Sigma$$

• In M^0 : $t^a = (\frac{\partial}{\partial t})^a$, take foliating Hypersurfaces $t = \text{const}$, $n^a = t^a = (1, 0, 0, 0)$

$$\Rightarrow T_{ab} t^a n^b = T_{00} \quad \text{and}$$

$$E = \int T_{00} d^3x$$

~~Φ : Gravitational Potential per unit Mass~~

* There is no known meaningful notion of the energy density of the gravitational field in G.R.

Newtonian: $\rho_E \sim E/V$ of a gravitational field

$$\rho_{\text{grav}} \sim - (8\pi)^{-1} |\nabla\phi|^2$$

• ϕ : is a metric component

$$\nabla^2 \phi = 4\pi\rho$$

$\hookrightarrow$ We expect the energy density in G.R. to be quadratic in the first derivatives of g_{ab} .

$$dU = \frac{1}{2} \rho \phi dV$$

$\hookrightarrow \frac{1}{2}$ avoids over counting and

$$E_{\text{grav}} = E_{\text{grav}} \phi \Rightarrow \left(\sum_i E_i \right) dV \frac{1}{2} \sim dU$$

$$U = \frac{1}{2} \int \rho \phi d^3x = \frac{1}{2} \int \left(\frac{\nabla^2 \phi}{4\pi} \right) d^3x$$

(Assuming ϕ and ρ have compact support)

$$U = -\frac{1}{8\pi} \int |\nabla\phi|^2 d^3x$$

• Isolated systems $\phi \sim \frac{1}{r}$

$$\Rightarrow U = \int \tilde{\rho}_{\text{grav}} d^3x$$

$$\tilde{\rho}_{\text{grav}} = -\frac{1}{8\pi} |\nabla\phi|^2$$

$$\vec{g} = -\nabla\phi$$

$$\vec{F} = -m\vec{g} = m\nabla\phi = \nabla(m\phi) \quad U = m\phi$$

* There does exist a Total energy of an isolated system: V^a present at $\mathcal{I}^\pm U^i$

* Special Relativity: Energy-momentum 4-vect.

$$P^\mu = (E, \vec{P})$$

ξ^μ : Time translation Killing field

$$\nabla_a \xi_b + \nabla_b \xi_a = 0$$

$$\nabla^a \xi_a = 0 = R^{ab} \partial_a \xi_b = 0$$

$$\nabla^2 \xi_a = 0$$

$$\Rightarrow -1 \partial_t \xi_t + \partial_i E_i = 0$$

$$i \xi^\mu \partial_\mu$$

$$e^{X^\nu} = X^\nu + i \xi^\nu$$

$$= X^\nu + i (\hat{t}, 0, 0, 0)$$

$$\hat{t} (-\partial_t \hat{t} + \sum \partial_i \hat{t}) = 0$$

$$-\partial_t \hat{t} - i (P_x + P_y + P_z) \hat{t} = 0$$

$$\int \frac{d\hat{t}}{\hat{t}} = \int -|\vec{P}|^2 dt$$

$$\hat{t} = A e^{-\int |\vec{P}|^2 dt}$$

$$\hat{t}(t, \vec{x}) = \int \frac{d^3p}{(2\pi)^3} e^{i\vec{p}\cdot\vec{x}} e^{-\int |\vec{p}| dt}$$

$$= A \int \frac{d^3p}{(2\pi)^3} e^{-(p_{xt} - i) t} e^{i\vec{p}\cdot\vec{x}}$$

$$\nabla^2 \xi_a = 0 \Rightarrow \nabla^2 \hat{t} = 0$$

$$\nabla_t \xi_t = 0 \dots$$

$$E \equiv -P_\mu \xi^\mu$$

$$\xi^\mu = (1, 0, 0, 0)$$

$$-M^2 = P^2$$

at rest

$$P^2 = 0^2 \Rightarrow P_{i,2} = 0^2 \leftrightarrow E = M$$

"Rest frame" Killing vector: $\xi^\mu = (1, 0, 0, 0)$
This is the rest frame for a massive
object: $SO(3)$ invariance

* Test bodies in Schwarzschild in orbits $r \gg r_s$
behave like test bodies in Newtonian grav
 $\rightarrow M$: Total mass of the spacetime

